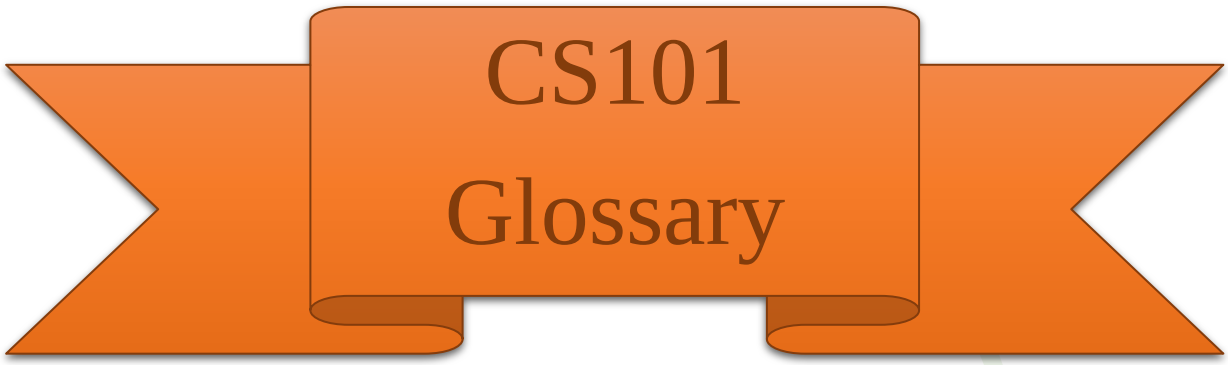


- **HTML**

Hypertext Markup Language As its name suggests, HTML is a collection of formatting commands that create hypertext documents--Web pages, to be exact. When you point your Web browser to a URL, the browser interprets the HTML commands embedded in the page and uses them to format the page's text and graphic elements. HTML commands cover many types of text formatting (bold and italic text, lists, headline fonts in various sizes, and so on), and also have the ability to include graphics and other nontext elements. Usually, html commands (tags) come in pairs depicting the start and end of a style; e.g. <B> This is bold text </B> would appear as bold text within a browser window. Development and maintenance of HTML standards is coordinated by the World Wide Web Consortium. See also: browser, HTML 3.0, hypertext, URL, VRML

- **100BaseT**

Another term for fast Ethernet, an upgraded standard for connecting computers into a local area network (LAN). 100BaseT Ethernet works just like regular Ethernet except that it can transfer data at a peak rate of 100 mbps. It's also more expensive and less common than its slower 10BaseT sibling. See also: fast Ethernet, Ethernet (10BaseT)

- **10BaseT**

The most common form of Ethernet is called 10BaseT, which denotes a peak transmission speed of 10 mbps using copper twisted-pair cable. Ethernet is a standard for connecting computers into a local area network (LAN). See also: Ethernet, fast Ethernet (100BaseT), hub

- **anonymous FTP**

Part of the Internet's appeal is the huge number of files available for download. Anonymous File Transfer Protocol (FTP) enables anyone to download files from a properly configured FTP server. This allows users without accounts to access files by entering the username anonymous, along with their email address as a password. Because anyone can access files that have been set up for anonymous FTP, Webmasters need to be careful to use it only for files meant for public consumption. See also: FTP

- **ANSI**

American National Standards Institute This organization of American industry groups works with the standards committees of other nations to develop standards to facilitate international trade and telecommunications. Some of ANSI's greatest hits in the computing field include ASCII, SCSI, and the ANSI.SYS device driver. See also: ASCII, SCSI

- **API**

application programming interface An API is a series of functions that programs can use to make the operating system do their dirty work. Using Windows APIs, for example, a program can open windows, files, and message boxes--as well as perform more complicated tasks--by passing a single instruction. Windows has several classes of APIs that deal with telephony, messaging, and other issues.

- **applet**

Applet is a diminutive form of app (application), and it refers to simple, single-function programs that often ship with a larger product. Programs such as Windows' Calculator, File Manager, and Notepad are examples of applets.

- **ARPAnet**

In the late 1960s--when the U.S. military was desperately afraid of a nuclear attack from the Soviet Union--some government computer scientists in the Advanced Research Projects Agency got together to design a bomb-proof network that would connect military bases and other military agencies. To do so, they created a system based on linking distant computers via a newly developed set of protocols called TCP/IP. This new nuke-proof network (though it was never tested with real nukes, of course) became ARPAnet. In the early 1980s, ARPAnet technology was put to use for nonmilitary purposes and gradually became what we now call the Internet. ARPAnet was taken out of commission in favor of a higher-speed network called NSFNET in 1990. See also: TCP/IP

- **ASCII**

American Standard Code for Information Interchange Bland, unformatted text files are best saved in ASCII (pronounced "askee") format. But ASCII is more than a text file format--it's a standard developed by the American National Standards Institute (ANSI) to define how computers write and read characters. The ASCII set of 128 characters includes letters, numbers, punctuation, and control codes (such as a character that marks the end of a line). Each letter or other character is represented by a number: an uppercase A, for example, is the number 65, and a lowercase z is the number 122. Most operating systems use the ASCII standard, except for Windows NT, which uses the suitably larger and newer Unicode standard. See also: ANSI

- **bandwidth**

In a general sense, this term describes information-carrying capacity. It can apply to telephone or network wiring as well as system buses, radio frequency signals, and monitors. On a more human level, the term can describe a person's capacity for dealing with multiple projects ("I'd like to update this database, but I don't have the bandwidth."). Bandwidth is most accurately measured in cycles per second, or hertz (Hz), which is the difference between the lowest and highest

frequencies transmitted. But it's also common to use bits or bytes per second instead. See also: memory bandwidth, video bandwidth

- **BASIC**

beginner's all-purpose symbolic instruction code This standard, high-level family of programming languages is simple to learn but creates programs that are typically slow to use. Many types of BASIC are available--your package might be called Turbo, Quick, or Visual Basic. The language is not generally used for industrial-strength applications, although Visual Basic has spawned a lot of shareware programs--and even some commercial apps such as MicroHelp's Uninstaller. See also: Visual Basic

- **beta software**

Beta versions of commercial software are work-in-progress test copies released prior to the full version. They're used to put the product through real-world tests and to ferret out bugs before the finished software hits the shelves. Betas often expire after a period of time, usually when the full version or the next beta is released. Originally, beta software was released only to developers, but increasingly betas are offered to the general public, usually through the manufacturer's Web site. Testing beta software can be a great way to try out a new product that you're not sure you want to buy. But be warned: if you decide to experiment with betas, expect an occasional software glitch or system crash. See also: demo software

- **binary**

A numeric system that represents all numbers using only two digits: 1 and 0. Binary is used by computers because each binary digit can be represented by a single digital switch set to either on or off. Each of these digital switches is called a bit. An 8-bit computer processor represents data using sets of 8 bits, a 16-bit processor uses 16 bits, and so on. The more bits a computer can process at once, the larger the numbers it can work with. See also: bit

- **bitwise operators**

JavaScript bitwise operators (&, |, ^, ~, <<, >>, >>>) work on numbers, just like arithmetic operators (+, -, /, *) do. But whereas arithmetic operators work on the actual values of numbers, bitwise operators work on their binary representations. Bitwise operators treat their operands as a set of bits (zeros and ones) rather than as decimal, hexadecimal, or octal numbers. Although the operators act upon binary representations of the operands, they return standard JavaScript numerical values. See also: bit, binary

- **BIOS**

basic input/output system The BIOS is what's coded into a PC's ROM to provide the basic instructions for controlling system hardware. The operating system and application programs both directly access BIOS routines to provide better compatibility for such functions as screen display. Some makers of add-in boards such as graphics accelerator cards provide their own BIOS modules that work in conjunction with (or replace) the BIOS on the system's motherboard. See also: ROM

❖ **bit**

binary digit A bit is the smallest unit of data in computing, with a value of either 0 or 1. Whenever you see a lowercase b associated with a number, it's likely to be a bit. It can be prefixed with kilo- (for 1,024 bits, or 2 to the 10th power) or mega- (1,024 x 1,024 bits)--and sometimes finds its way into data transfer speeds (such as 14.4 kbps). See also: byte, gigabyte, kbps, megabyte

❖ **Boolean**

English mathematician George Boole (1815-64) founded a field of mathematical and philosophical study called symbolic logic. His name is now used most often to describe a subset of symbolic logic: constructing database queries. Whenever you see a Web search tool or database query system that allows you to use AND, OR, and NOT to hone your search, the chances are it uses Boolean techniques. The most common Boolean operators are AND (you're looking for all terms), OR (you're looking for at least one of the terms), and NOT (you're excluding a term). You'll always see the operators referred to in uppercase letters, although you usually don't need to enter them that way to make a Boolean search work properly. Also, the Boolean operator AND doesn't work like a normal English and. For example, a Boolean search through a database of rock musicians for members of the Beatles AND Wings would turn up only Paul McCartney. For details on how to construct Boolean queries, read CNET's Boolean search primer. See also: Boolean operator

❖ **Boolean operator**

This is one of a number of conjunctions (AND, OR, NOT, and less commonly WITH or ADJ) used in constructing Boolean queries. For details on how to construct such queries, read CNET's Boolean search primer. See also: Boolean

❖ **browser**

If you can read this, it's highly likely that you're using a Web browser. In brief, a browser is your interface to the World Wide Web; it interprets hypertext links and lets you view sites and navigate from one Internet node to another. Among the companies that produce browsers are NCSA Mosaic, Netscape, and Microsoft, as well as commercial services like CompuServe, Prodigy, and America Online. See also: hypertext

❖ **bus**

In broadest terms, a bus is a common connection between electrical devices. In computerese, bus most commonly means the data pathway that connects a processor to memory and to other "peripheral" buses, such as VESA and PCI. See also: PCI, VESA

❖ **byte**

A byte usually denotes 8 bits (also called an octet or a word), which the computer treats as a single unit. Longer sequences like 16 and 32 bits are also possible. Abbreviated as uppercase B. See also: bit, gigabyte, megabyte.

❖ **bytecodes**

A Java compiler creates platform-independent bytecodes that run inside of a Java Virtual Machine (VM). That means that a Java applet can execute on any machine that supports a Java VM.

- **cable modem**

Speed is something we all want on the Internet, and it also explains the allure of cable modems, which promise speeds of up to 80 times faster than an ISDN line or 6 times faster than a dedicated T1 line (the type of connection most large corporations use). Because cable modems provide Internet access over cable TV networks (which rely primarily on fiber-optic or coaxial cable), they are much faster than modems that use phone lines. But cable modems are expensive--they're expected to cost between $200 and $250 when they hit the retail shelves in mid-1998. And the fees from your local cable company could include a $50 to $100 installation fee, while the service itself costs around $40 per month, if the service is offered at all. Some cable modem testers complain that the service is neither as fast nor as consistent as promised. On the bright side, in March 1997, the Multimedia Cable Network Systems (MCNS) released cable modem specifications in an attempt to establish standards, which will guarantee interoperability between different manufacturers' modems and should reduce cable modem prices due to increased competition. And some cable companies may rent the modems to consumers, which could give you the opportunity to test out the service before you make a purchase. See also: modem, ISDN

- **cache**

Caches come in many types, but they all work the same way: they store information where you can get to it fast. A Web browser cache stores the pages, graphics, sounds, and URLs of online places you visit on your hard drive; that way, when you go back to the page, everything doesn't have to be downloaded all over again. Since disk access is much faster than Internet access, this speeds things up. Of course, disk access is slower than RAM access, so there's also disk caching, which stores information you might need from your hard disk in faster RAM. See also: L1 cache, L2 cache, primary cache, secondary cache

- **CSS**

Cascading Style Sheets Cascading Style Sheets are a big breakthrough in Web design because they allow developers to control the style and layout of multiple Web pages all at once. Before Cascading Style Sheets, changing an element that appeared on many pages required changing it on each individual page. Cascading Style Sheets work just like a template, allowing Web developers to define a style for an HTML element and then apply it to as many Web pages as they'd like. With CSS, when you want to make a change, you simply change the style, and that element is updated automatically wherever it appears within the site. Both Navigator 4.0 and Internet Explorer 4.0 support Cascading Style Sheets. If you needed any more proof of the problem-solving nature of CSS, the World Wide Web Consortium (W3C) has recommended Cascading Style Sheets (level 1) as an industry standard. See also: DHTML, HTML

- **CCITT**

Comite Consultatif Internationale de Telegraphie et Telephonie This is an international committee based in Geneva, Switzerland, that recommends telecommunications standards, including the

audio compression/decompression standards (codecs) and the famous V. standards for modem speed and compression (V.34 and so on). Although this organization changed its name to ITU-T (International Telecommunications Union-Telecommunication), the old French name lives on.

- **CD-R**

compact disc recordable A CD-ROM format that enables you to record data onto compact discs so that regular CD-ROM drives can read it. With a CD-R drive, you can record data onto a recordable disc on different occasions, known to experts as multiple sessions.

- **CD-ROM**

compact disc, read-only memory A compact disc used to store and play back computer data instead of digital audio. CD-ROMs can contain up to 650MB of data (though they often contain a lot less). CD-ROMs have become a favorite medium for installing programs, since they cost only slightly more to manufacture than floppy disks, and most major software applications come on at least five floppies. Don't sound hopelessly out of touch with technology: use the term CD-ROM to refer to the technology or the discs, but not to the hardware you play the discs on. That's a CD-ROM drive. See also: CD-ROM drive

- **CD-ROM drive**

A computer peripheral that plays back CD-ROMs and--with the right software--audio CDs. It consists of a spindle that revs up the disc, a laser that flashes onto the disc's uneven surface, a prism that deflects the laser beam, and a light-sensitive diode that reads the flashing light. Since the audio CD standard calls for data transfer of 150 kilobytes per second (KB/sec), all CD-ROM drives can handle this speed, and most can manage 2X (double-speed, or 300 KB/sec) or 4X (quad-speed, or 600 KB/sec) rates. There are also 6X (900 KB/sec) and even 8X (1,200 KB/sec) drives, but since most CD-ROM titles are squarely aimed at transferring data within the 2X/4X bandwidth, you'll usually not see much advantage to drives with higher transfer rates than with a 4X drive. See also: CD-ROM

- **CD-ROM XA**

CD-ROM, extended architecture This CD-ROM standard holds the unique distinction of having been a must-have checklist item for CD-ROM drives before it even became a standard. While the standards committee was hashing out how "extended" the new architecture should be, every CD-ROM drive maker was slapping an "XA-ready" label on its boxes. Most of the time, this was the manufacturers' way of saying that the drives read multisession CD-ROMs.

- **CGI**

common gateway interface The CGI standard lays down the rules for running external programs in a Web HTTP server. External programs are called gateways because they open up an outside world of information to the server. See also: HTTP

- **cgi-bin**

CGI binary When you see this directory in your Web browser's URL window, it's a sign that you're running a CGI program, such as a search tool. See also: CGI

- **chat**

This rather generic term has come to describe one of the more popular activities on the Internet. Using special software, Internet users can enter chat areas or "virtual spaces," where they can communicate in real time. While most chat software only lets users talk by typing, more advanced products assign avatars, 2D or 3D characters, to each participant. These avatars may even have expressions selected by the chatters. The most advanced products not only use avatars, but also let users with sound cards speak to each other.

- **CMOS**

complementary metal-oxide semiconductor Chips made with this low-power semiconductor technology are commonly used in personal computers to hold basic start-up information--such as the time and date--for use by the system's BIOS. See also: BIOS

- **coaxial cable**

Typically used to connect a television to cable TV services, coaxial cable consists of a small copper tube or wire surrounded by an insulating material and another conductor with a larger diameter, usually in the form of a tube or copper braid. This cable is then encased in a rubberized protective material. See also: modem, HFC network, modulation, demodulation

- **COBOL**

common business-oriented language Developed in 1960 by a team led by the National Bureau of Standards (now the National Institute of Standards and Technology, or NIST), COBOL was the first standardized business computer-programming language. In fact, that's what COBOL stands for: common business-oriented language. Since COBOL was intended for business use, many functions, such as payroll and accounting, are still executed using programs written in the language

- **codec**

coder/decoder or compression/decompression algorithm As the name implies, codecs are used to encode and decode (or compress and decompress) various types of data--particularly those that would otherwise use up inordinate amounts of disk space, such as sound and video files. Common codecs include those for converting analog video signals into compressed video files (such as MPEG) or analog sound signals into into digitized sound (such as RealAudio). Codecs can be used with either streaming (live video or audio) or files-based (AVI, WAV) content.

- **.com**

1. A file extension used for PC programs, especially those that run under DOS (for example, command.com, win.com). 2. A domain type used for Internet locations that are part of a business or commercial enterprise (for example, www.cnet.com). See also: domain name, file extension

- **COM port**

Although it's in all capital letters, COM is not an acronym. It's a contraction of communications, and it's used to describe the serial port on a PC. COM is generally used in conjunction with a number, as in COM1, COM2, COM3, or COM4.

- **command-line interface**

Before graphical interfaces like Windows, the Mac OS, and X, users interacted with their personal computers by typing text commands. While the drag-and-drop simplicity of a GUI is attractive to many users, particularly newcomers, command-line interfaces often allow greater flexibility and control and are therefore still preferred by many power users. see also: GUI

- **cookie**

According to Netscape, cookies are a "general mechanism which server side connections can use to both store and retrieve information on the client side of the connection." In English, that means cookies are small data files written to your hard drive by some Web sites when you view them in your browser. These data files contain information the site can use to track such things as passwords, lists of pages you've visited, and the date when you last looked at a certain page.

- **CPU**

central processing unit The CPU--a highly complex silicon chip ranging from the size of a matchbook to the wallet-sized Intel Xeon--is your computer's brain, taking requests from applications and then processing, or executing, actions, a.k.a. operations. The faster your processor, the more operations it can execute per second. The more operations you have per second, the faster things happen in your applications; thus, games play more smoothly, and spreadsheets calculate more quickly. Sometimes the term CPU is also used to describe the whole box that contains the chip (along with the motherboard, expansion cards, disk drives, power supply, and so on). Both uses are widespread, but only the first is really accurate. See also: core logic, FPU

- **cps**

characters per second Once commonly used to describe the relative speed of printers, cps now sometimes refers to the data transfer rates of modems. For practical purposes, on the PC, a character is equal to eight bits or one byte. See also: bit, bps, byte

- **core logic**

The central functions of a computer are usually handled by a set of chips called the core logic chipset. These functions are the system's traffic cops: a cache for instructions, a controller for handling memory, bus interface logic, and data path functions. See also: CPU

- **CRC**

cyclical redundancy check CRC is a mathematical technique used to check for errors when sending data by modem. Because some phone lines are notoriously crackly and can cause breaks in transmission, this is a crucial step. If the CRC fails to add up, the receiving end of a data

transmission sends a NAK (negative acknowledgement or "say that again") signal until it does add up. CRCs are also used in tape backups and other streaming communications.

- **CRT**

cathode ray tube The active component of monitors and TVs, the cathode ray tube is a big bell of glass with electron guns at one end and a viewing screen at the other.

- **cyber-**

The prefix cyber- is most often used to make whatever word it's attached to seem hip, cool, and connected in some loose way to the world of computers or the Internet. The habit probably started with science fiction writer William Gibson, who coined the term cyberspace. See also: cyberspace

- **cyberspace**

Science fiction writer William Gibson coined the term cyberspace in the perennial favorite novel Neuromancer. Gibson used the word to describe a virtual world of computer networks that his cyberpunk heroes "jacked into." Everyone else uses the word cyberspace loosely to refer to virtual reality, the Internet, the World Wide Web, and many other kinds of computer systems that users become immersed in. It's about as ill-defined a term as information superhighway, but it's much hipper. See also: information superhighway

- **daemon**

Pronounced "demon," this Unix term refers to a specific type of program or agent designed to work in the background. Using a daemon, a program can simply hand off data to the smaller program and go on to more important things. For example, a print daemon could handle print requests from multiple users and applications, freeing them for other tasks.

- **database**

A database can be as simple as a shopping list or as complex as a collection of thousands of sounds, graphics, and related text files. Database software is designed to help users organize such information. While early "flat" databases were limited to simple, searchable rows and columns, modern relational databases allow users to access and reorganize data in a variety of ways. Even more advanced databases let users store and retrieve all kinds of nonstandard data, from sound clips to video.

- **device driver**

A program that lets peripheral devices communicate with computers. Some device drivers for standard components, such as keyboards or monitors, come with computers. Devices that are added later require the user to install the corresponding drivers

- **DHCP**

Dynamic Host Configuration Protocol DHCP is a protocol for dynamically assigning IP addresses to networked computers. With DHCP, a computer can automatically be given a unique IP address each time it connects to a network--making IP address management an easier task for network

administrators. When a computer logs on to the network, the DHCP server selects an IP address from a master list and assigns it to the system. Windows 95 and NT systems include DHCP support.

❖ **DHTML**

dynamic HTML The W3C may not have an official definition for it, but dynamic HTML combines HTML, style sheets, and scripts to make Web pages more interactive. It sounds great. But while both Microsoft Internet Explorer 4.0 and Netscape Communicator 4.0 support dynamic HTML features, they do so in different--and incompatible--ways. The companies say they will work to make their browsers compatible with whatever specification the W3C finally decides upon. However, the browsers may be in their fifth generation before users see a concrete standard. See also: CSS, HTML

❖ **digital certificate**

Citing concerns about security, many people are still wary of online transactions. In an attempt to assuage those fears, software vendors, security specialists, and online vendors have developed the concept of digital certificates. A digital certificate is a password-protected file that includes a variety of information: the name and email address of the certificate holder, an encryption key that can be used to verify the digital signature of the holder, the name of the company issuing the certificate, and the period during which the certificate is valid. Certificate authorities (CAs) gather information about a person or company and then issue certificates. These certificates can be used as online identification, much in the same way a driver's license can verify your identity in the physical world. If an email message or order form comes through with an attached digital certificate, the recipient can be more confident that the document is genuine. Several technologies (including SET, SSL, and Authenticode) are currently competing for market share, each hoping to become the certificate of choice. See also: encryption

digital signature

Forgery is a growing concern among Netizens. After all, who's to say that a message with your name on it is really from you and not somebody pretending to be you? Digital signatures are a means of proving that a file or email message belongs to a specific person, much as a driver's license proves identity in real life. Digital signatures have the added benefit of verifying that your message has not been tampered with. When you sign a message, a hash function--a computation that leaves a specific code, or "digital fingerprint"--is applied to it. If the fingerprint on the recipient's message doesn't match the original fingerprint, the message has been altered. Digital signatures are often used in combination with strong-encryption software to create a secure channel of communication, in which both privacy and identity are protected. See also: digital certificate, encryption

❖ **DIMM**

dual in-line memory module As you'd expect from its name, a DIMM has a lot more bandwidth than a single in-line memory module (SIMM). It's a small circuit board filled with RAM chips, and its data path is 128 bits wide, making it up to 10 percent faster than a SIMM. DIMMs are

prevalent on the Power Mac platform but are also creeping into high-performance systems. See also: SIMM, RAM

- **diode**

These semiconductive electronic components allow current to flow through them in only one direction. Most people are familiar with light-emitting diodes (LEDs), which produce light when current is applied, and liquid-crystal diodes (LCDs), which lighten or darken a transparent material when current is applied. LCDs are often used to create flat-panel displays for portable computers. see also: flat-panel display, LCD, semiconductor

- **DLL**

dynamic link library When you're trying to share functions or resources among Windows programs, you generally put them in DLLs. These libraries do all kinds of things, from holding icons (Windows 3.1's moricons.dll) to enabling Visual Basic programs (vbrun3.dll) to creating network sockets that allow you to hook up to the Internet (winsock.dll).

- **DMA**

direct memory access Hardware devices attached to PCs (ranging from keyboards to sound cards) can be designed to send their instructions to and from main memory in one of two ways. The default is to ask the CPU to do the work. The more efficient way is to allocate one of the PC's DMA channels to send instructions directly to memory. This leaves the CPU free to do more important things. Like IRQs, DMA channels are limited in number, and you can't allocate one channel to more than one device (unless you want to grind your system to a halt). Most users come in contact with DMA when they install a sound card that--if they're lucky--picks the right channel during setup. See also: Plug and Play

- **DNS**

domain name system When you send email or point a browser to an Internet domain such as cnet.com, the domain name system translates the names into Internet addresses (a series of numbers looking something like this: 123.123.23.2). The term refers to two things: the conventions for naming hosts and the way the names are handled across the Internet

- **domain name**

Looking for a domain name? You'll find it to the right of the @ sign in an email address, or about ten characters into a URL. VU's domain name is vu.edu.pk The domain name of the Ministry of Science & Technology is most.gov.pk International domain names are issued by the National Science Foundation (NSF), and they come with different extensions based on whether the domain belongs to a commerical enterprise (.com), an educational establishment (.edu), a government body (.gov), the military (.mil), a network (.net), or a nonprofit organization (.org). Some domains use a geographical notation too (such as the San Francisco, California-based well.sf.ca.us). Country specific domain names (such as those with the .pk suffix) are assigned by other agencies.

- **dot matrix printer**

One of the earliest mass-market attempts at reliable, high-speed printing, dot matrix printers used a set of pins that could be fired rapidly at an inked ribbon to form characters on an underlying sheet of paper. While speedy, these printers suffered from relatively poor print quality. Even higher-end models with as many as 48 pins could produce only near-letter-quality text. Lower-end, 9-pin models were suitable only for printing drafts.

- **dpi**

dots per inch A measure of resolution in printers and scanners, dpi ratings describe how many pixels such devices can fit into an inch of space. A 400-dpi printer, for example, produces sharper printouts than a 300-dpi device. See also: resolution

- **DRAM**

dynamic RAM DRAM is the Hershey's chocolate of readable/writable memory: it's not the best, but it's cheap, does the job, and is available almost everywhere you look. DRAM data resides in a cell made of a capacitor and a transistor. The capacitor tends to lose data unless it's recharged every couple of milliseconds, and this recharging tends to slow down the performance of DRAM compared to speedier RAM types. See also: EDO RAM, SRAM, RAM

- **DSL**

digital subscriber line Digital subscriber lines carry data at high speeds over standard copper telephone wires. With DSL, data can be delivered at a rate of 1.5 mbps (around 30 times faster than through a 56-kbps modem). Also, DSL users can receive voice and data simultaneously, so small offices can leave computers plugged into the Net without interrupting phone connections. Currently, DSL is expensive because specialized equipment--a splitter--needs to be installed at the subscriber's location. DSL Lite, the consumer-ready version of DSL, requires no such splitter, and promises comparable access speeds at a cheaper rate. xDSL is the collective term for different variations of DSL, such as ADSL and HDSL. PTCL has plans to offer DSL lines in Pakistan.

- **duplex**

If you have a printer that can print on both sides of a sheet of paper, it's duplex. If you have a communications line that lets you send and receive data (or talk and listen) at the same time, it's duplex. If you use a cheap speakerphone, you'll notice that as soon as it picks up any noise at your end, it cuts off whatever was being said by the person on the other end. That speakerphone is not duplex. See also: half duplex, full duplex

- **DVD**

digital versatile disc Originally referred to as digital video discs, these high-capacity optical discs are now used to store everything from massive computer applications to full-length movies. While similar in physical size and appearance to a compact disc or a CD-ROM, DVD is a huge leap from its predecessor's 650MB of storage. A standard single-layer, single-sided DVD can store a whopping 4.7GB of data. But it doesn't stop there--DVD also has a two-layer standard that boosts the single-sided capacity to 8.5GB. And there's more! DVDs can be double-sided, ramping up the maximum storage on a single disc to 17GB. Unfortunately, to use DVDs, you'll have to buy a new

drive, but that new hardware will also read your older CD-ROMs and audio CDs. see also: CD-ROM

- **EDO RAM**

extended data-out RAM This form of dynamic RAM speeds access to memory locations by working on a simple assumption: the next time memory is accessed, it will be at a contiguous address in a contiguous chunk of hardware. This assumption speeds up memory access times by up to 10 percent over standard DRAM.

- **EEPROM**

electrically erasable programmable ROM With normal ROMs you have to replace the chip (or chips) when new BIOS instructions are introduced. With EEPROMs, a program tells the chip's controller to give it electronic amnesia and then downloads the new BIOS code into it. This means a manufacturer can easily distribute BIOS updates on floppy, for instance. This feature is also called flash BIOS, and you might also come across it in devices like modems and graphics/video cards. See also: BIOS, ROM

- **EIDE**

enhanced integrated device (or drive) electronics A second take on the PC platform's IDE drive interface that increases the previous maximum disk size from 504MB to more than 8GB, speeds up the data transfer rate to more than twice what IDE was capable of, and doubles the number of drives a PC can contain, bringing the number up to four. On the PC platform, EIDE gives SCSI-2 a run for its money, and while most people agree that SCSI-2 is technically superior, EIDE is cheaper to implement, which gained it widespread acceptance. See also: DRAM, EDO RAM, IDE, RAM

- **EISA bus**

Enhanced Industry-Standard Architecture bus A 32-bit bus for PCs built around 386, 486, or Pentium chips. EISA was developed as an alternative to IBM's microchannel bus, and is more compatible with the original ISA bus. EISA computers can generally use ISA cards as well. The use of the EISA bus is declining after the introduction of the PCI bus. See also: bus, ISA, PCI

- **email**

electronic mail Whenever you send messages to people using a computer and they read it later, you've sent a piece of email. You can send email in several ways--across a local area network, via the Internet, or through an online service like CompuServe or America Online--and you can send it to a single recipient or to a whole slew of them. But all email behaves pretty much the same way: you send it to a virtual mailbox, and the recipient has to pick it up or can use software that does it automatically. The VU LMS has a built in email system for its students. See also: snail mail, spam

- **emoticon**

The trouble with online chat, postings, and email is that anything said in them is open to misinterpretation. Emoticons are clusters of punctuation such as :) and : > that are used to set the

tone for the sentence that precedes them. The knack to interpreting them is to tilt your head sideways to the left and look for a facial expression. Here's a brief rundown of the less ridiculous emoticons: :) a grin :-) another grin :-(a frown ;-) a wink : > a wicked grin @--->--->- a long-stemmed rose

❖ **encryption**

Encryption is the process of changing data into a form that can be read only by the intended receiver. To decipher the message, the receiver of the encrypted data must have the proper decryption key. In traditional encryption schemes, the sender and the receiver use the same key to encrypt and decrypt data. Public-key encryption schemes use two keys: a public key, which anyone may use, and a corresponding private key, which is possessed only by the person who created it. With this method, anyone may send a message encrypted with the owner's public key, but only the owner has the private key necessary to decrypt it. PGP (Pretty Good Privacy) and DES (data encryption standard) are two of the most popular public-key encryption schemes

❖ **EPP**

Enhanced Parallel Port Developed by Intel, Xircom, Zenith, and other companies, the EPP spec builds on the design of the humble parallel port, adding two-way communications and faster exchange. It's been used in laptops since mid-1991, where it has been in great demand because of the proliferation of hard disks, tape backup units, CD-ROM drives, and other mass-storage devices that use parallel ports. Since both EPP and ECP are supported by the IEEE 1284 specification, many new parallel ports can run in either fashion. See also: ECP, parallel port

❖ **EPROM**

erasable programmable ROM Unlike dynamic and static RAM, EPROM chips do not require power to retain their data. Consequently, EPROM chips are commonly used to store BIOS information and basic software for modems, video cards, and other peripherals. EPROMs can be erased by exposure to ultraviolet light and then reprogrammed using a device called a PROM burner. See also: BIOS, DRAM, EEPROM, RAM, ROM, SRAM

❖ **Ethernet**

Ethernet is a standard for connecting computers into a local area network (LAN). The most common form of Ethernet is called 10BaseT, which denotes a peak transmission speed of 10 mbps using copper twisted-pair cable. See also: fast Ethernet (100BaseT), hub

❖ **extranet**

Companies often use extranets to provide nonpublic information to a select group of people, such as business partners or customers. So while an extranet may look like an ordinary Web site, you have to enter a password or use digital encryption to access it. For example, Federal Express's customers can track packages on the company's extranet by simply entering a tracking number. And Bank of America's extranet lets users transfer funds or look up account balances online. Using an extranet can help companies save money by allowing customers to find information themselves, without having to call and talk to a person. See also: intranet.

❖ **FAQs**

frequently asked questions These text files are supposed to answer all the questions a newcomer to an online site might have. Designed to cut down on basic tech support queries, FAQs can be organized in virtually any structure, and they often cover a far wider range of subjects than just basic site orientation. You can either pronounce it "facks" or sound out each letter.

❖ **fast Ethernet**

Fast Ethernet is an upgraded standard for connecting computers into a local area network (LAN). It works just like regular Ethernet except that it can transfer data at a peak rate of 100 mbps. Also referred to as 100BaseT, fast Ethernet is more expensive and less common than its slower 10BaseT sibling. See also: Ethernet (10BaseT)

❖ **FAT**

file allocation table PCs store and retrieve files on hard disks using this filing system, which divides the hard disk into "clusters" of bytes and then files data into these clusters. When a program calls for a file, the FAT looks up the locations of all the clusters where the data is stored. The cluster size depends on the size of the hard disk, with bigger drives using bigger clusters. Here's the tricky part: a single cluster can only store data from a single application or file; if the data doesn't fill the whole cluster, then the rest of that hard disk drive space is wasted. Let's say a cluster is 32K; a 42K word processing document would take up a full cluster, plus 10K of a second. The 22K remaining in the second cluster remain unused, no matter how full the hard disk appears to be.

❖ **fiber-optic cable**

Fiber-optic cables consist of thin filaments of glass (or other transparent materials), which can carry beams of light. A laser transmitter encodes frequency signals into pulses of light and sends them down the optical fiber to a receiver, which translates the light signals back into frequencies. Less susceptible to noise and interference than other kinds of cables, optical fibers can transmit data greater distances without amplification. But because the glass filaments are fragile, optical fiber must be run underground rather than overhead on telephone poles.

❖ **file extension**

In filenames, the group of letters after the period is called the file extension. For example, if the filename is command.com, the extension is .com. Typically, under DOS/Windows operating systems, executable programs have extensions of .exe .com .bat The Unix operating system family does not require specific extensions for its executable programs. See also: .com

❖ **FPU**

floating-point unit Your computer's CPU is geared toward dealing with integer mathematics. When you throw floating-point math at it, the CPU shucks off the responsibility to the FPU (if the computer has one), which is designed to handle floating-point math more efficiently. Once called a numeric (or math) coprocessor, the FPU can be either a separate chip (such as Intel's 80387 or Motorola's 68881), or it can be integrated into the CPU, such as the Pentium or 68040 processors.

Just as 1994's flawed Pentiums could be induced to perform floating-point operations (by running a program to disable the FPU), your CPU can, too--but it won't do it as quickly as an FPU. See also: CPU

- **firewall**

If you want to protect any networked server from damage (intentional or otherwise) by those who log in to it, you put up a firewall. This could be a dedicated computer equipped with security measures such as a dial-back feature, or it could be software-based protection called defensive coding.

- **flash ROM**

This jargon refers to ROM chips that can be reprogrammed with new BIOS instructions after the chips have left the factory. Such ROM chips are technically called EEPROMs

- **freeware**

Freeware is software you can download, pass around, and distribute without payment. However, it's still copyrighted, so you can't turn around and decompile it or sell it as your own (as you can with a public domain program). See also: public domain, shareware

- **FTP**

file transfer protocol This Internet protocol is used to copy files between computers--usually a client and an archive site. It's old-fashioned, it's a bit on the slow side, it doesn't support compression, and it uses cryptic Unix command parameters. But the good news is that you can download shareware or freeware apps that shield you from the complexities of Unix, and you can connect to FTP sites using a Web browser.

- **GPS**

Global Positioning System The Global Positioning System is a satellite navigation system used to determine terestrial position, velocity, and time. Once exclusively used by the U.S. military, the GPS is now available to the worldwide general public. The GPS system relays satellite signals that can be processed by a GPS receiver

- **GSM**

Groupe Speciale Mobile (Global System for Mobile Communications) This set of standards is widely used in Europe for cellular communications. The audio encoding subset of the GSM standard is best known to computer users because its data compression and decompression techniques are also being used for Web-phone communication and encoding WAV and AIFF files

- **GUI**

graphical user interface (pronounced gooey) A graphical user interface lets users interact with their computer via icons and a pointer instead of by typing in text at a command line. Popular GUIs, such as Sun Microsystem's OpenWindows, Microsoft's Windows, and Apple's Mac OS, have freed

many users from the command-line interfaces like MS-DOS and Unix. see also: command-line interface

- **hit**

Probably the most misused term in Net vocabulary, a hit can refer to any one of a few different things. If you perform a search using Excite or SHAREWARE.COM, the results are called hits. If you load up a Web page, you've hit the site. However, when people say, "Our Web site had 2,000 hits" (meaning that there were 2,000 visitors, or 1,000 visitors looking at two pages each), that's simply wrong. Technically, a hit is a request made to the Web server. For example, if you look at a Web page that contains ten GIF files, one person visiting one page will make 11 hits on the server: one for the page, and ten for the graphics on the page.

- **home page**

Web sites are by nature tangled groups of interconnected pages. To make them easier to navigate, the sites have one or more home pages that you can use for orientation. A home page serves as the site's introduction, starting point, and guide.

- **HTML 3.0**

This extension of Hypertext Markup Language expands the original HTML specification, which was pretty basic and allowed documents to contain only one column of text, a little formatting, and a few graphics. Developed by the World Wide Web Consortium and Web software developers, HTML 3.0 offers enhanced graphics capabilities and enables Web-page designers to put in tables and frames and to flow text around figures. To read documents written in HTML 3.0 you must have a browser that supports it. See also: HTML

- **HTTP**

hypertext transfer protocol The protocol used to transmit and receive all data over the World Wide Web. When you type a URL into your browser, you're actually sending an HTTP request to a Web server for a page of information (that's why URLs all begin with "http://"). HTTP1.1, the latest version, is currently undergoing revisions to make it work more efficiently with TCP/IP See also: protocol, URL, client, server, TCP/IP

- **hub**

This chunk of hardware is used to network computers together (usually over an Ethernet connection). It serves as a common wiring point so that information can flow through one central location to any other computer on the network. See also: Ethernet

- **heuristics**

A method of analyzing outcome through comparison to previously recognized patterns. For example, an antivirus program, familiar with behaviors typical of viruses (such as deleting files in sequence), could use heuristics to identify unknown virus strains by their behavior

- **hyperlink**

Hyperlinks are the easy-to-spot underlined words or phrases you click in World Wide Web documents to jump to another screen or page. Hyperlinks contain HTML-coded references that point to other Web pages, which your browser then jumps to. Also called anchors. See also: anchor

- **anchor**

Anchors are the same as hyperlinks--the underlined words or phrases you click on in World Wide Web documents to jump to another screen or page. The word anchor is used less often than hyperlink, but it does maintain the seafaring theme of navigating and surfing the Net. See also: hyperlink

- **hypermedia**

One step ahead of hypertext, hypermedia integrates text, images, video, and sound into its documents. The elements of hypermedia documents are linked in an interactive way. The best way to describe hypermedia is by example--and the most common examples are the sound-laden pages of the World Wide Web. See also: hyperlink, hypertext

- **hypertext**

Hypertext is a nonsequential way of presenting information. Hypertext links information in a complex web of associations, powered by hyperlinks. Essentially a way of browsing information, hypertext is a way to describe how you learn information from a well-designed CD-ROM encyclopedia or from the World Wide Web. See also: hyperlink, hypermedia

- **hexadecimal**

While the more common decimal system uses a base of ten to represent all possible numbers, hexadecimal notation uses a base of sixteen: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, A, B, C, D, E, F. While such a system may seem confusing at first, it works extremely well in computing environments. For example, a single byte of information can be represented as eight bits (10011101), decimal numerals (913), or simplified to hex (9D). In hex, every byte can be shown as two hexadecimal characters. As it relates to Web design, hexadecimal is the alphanumeric system used to specify colors in HTML. For example, the hexadecimal equivalent of white is FFFFFF, while black is 000000.

- **IDE**

integrated device (or drive) electronics This hard-drive interface has all of its controller electronics integrated into the drive itself. The IDE specification handles hard disks up to 504MB in size. Because of its simple instruction set and the short route between controller and drive, it's a quick and easy type of drive to use (unlike SCSI drives and the older Enhanced Small Device Interface (ESDI) drives from the late 1980s). But because it's a limited specification, IDE is gradually being superseded by an enhanced version, EIDE. See also EIDE, SCSI

- **IEEE**

Institute of Electrical and Electronics Engineers This nonprofit U.S. engineering organization develops, defines, and reviews standards within the electronics and computer science industries

❖ **IEEE 1394**

Also known as FireWire, IEEE 1394 is a standard for high-speed serial connections approved by the Institute of Electrical and Electronics Engineers. The standard is designed for the exchange of information between PCs and consumer electronics devices that transfer large amounts of data, such as digital camcorders or VCRs. Currently, there is an extremely limited number of FireWire devices, but proponents of PC and TV convergence think the standard will be more widely adopted as time goes on. Windows 98 will support FireWire-compliant devices

❖ **IMAP**

Internet Message Access Protocol IMAP provides a means of managing email messages on a remote server, similar to the POP protocol. But IMAP offers more options than POP, including the ability to download only message headers, create multiuser mailboxes, and build server-based storage folders. See also: POP

❖ **information superhighway**

This is a buzzword from a speech by U.S. Vice-President Al Gore that refers to the Clinton/Gore administration's plan to deregulate communication services and thus widen the scope of the Internet by opening carriers, such as television cable, to data communication. The term is widely and loosely used to mean the Internet, and it's often shortened to I-way, the infobahn, and so on.

❖ **Ink jet printer**

Unlike daisy wheel and dot matrix printers, which rely on physical contact with the an inked ribbon and paper, ink jet printers simply spray ink in a series of dots to form characters. While early ink jets were prone to clogging, they were capable of delivering good print speeds and relatively high-quality text and graphics. Today's best ink jet printers have eliminated most of the clogging problems, rival laser printers for text quality, and can also produce color images with near-photographic quality--all for even less money than a low-end laser printer. See also: daisy wheel printer, dot matrix printer, laser printer

❖ **Internet**

The Internet originated in 1969, in the midst of the Cold War, as a "nuke-proof" communications network. As you might guess, it received most of its early financing from the U.S. defense department. Now, however, it consists of countless networks and computers across the world that allow millions of people to share information. The lines that carry the majority of the information are know as the Internet backbone. While the government used to run things, now major Internet service providers (ISPs) such as MCI, GTE, Sprint, UUNET, and ANS own portions of the backbone--a good thing as they have the motivation and the revenue to maintain the quality of these large networks. For answers to commonly asked questions about the Internet, check out our recent feature: "20 questions: how the Net works." See also: intranet, extranet, browser, ARPAnet

❖ **intranet**

A play on the word Internet, an intranet is a restricted-access network that works like the Web, but isn't on it. Usually owned and managed by a corporation, an intranet enables a company to share its resources with its employees without confidential information being made available to everyone with Internet access.

- **I/O port**

input/output port Any socket in the back of a computer that you use to connect to another piece of hardware is called an I/O port.

- **IP**

Internet protocol The Internet protocol defines how information gets passed between systems across the Internet. See also: IP address, VAT

IP address

Internet protocol address This address is a unique string of numbers that identifies a computer on the Internet. These numbers are usually shown in groups separated by periods, like this: 123.123.23.2. All resources on the Internet must have an IP address--or else they're not on the Internet at all. See also: IP, VAT

- **IP packet**

An IP packet is a chunk of data transferred over the Internet using standard Internet protocol (IP). Each packet begins with a header containing addressing and system control information. Unlike uniform ATM "cells," IP packets vary in length depending on the data being transmitted. See also: IP, IP address, ATM

- **IRC**

Internet Relay Chat IRC is a way of hooking up with other Net users to exchange written comments--live and in real time. To do this, you need an IRC client and an IRC server. Once connected to the server, you join a channel, or discussion group, which can include people from all over the world. IRC channels may hold discussions about anything under the sun (and some topics that shouldn't see the light of day). IRC can be accessed by a variety of downloadable software on both the PC and Mac.

- **ISDN**

Integrated Services Digital Network The plain old telephone system doesn't handle large quantities of data, and the phone companies realized this a long time ago. So the ISDN spec was hammered out in 1984 to allow for wide-bandwidth digital transmission using the public switched telephone network. Under ISDN, a phone call can transfer 64 kilobits of digital data per second. But it's not always easy to adopt.

- **ISP**

Internet service provider Once upon a time, you could only connect to the Internet if you belonged to a major university or had a note from the Pentagon. Not anymore: ISPs have arrived to act as

your (ideally) user-friendly front end to all that the Internet offers. Most ISPs have a network of servers (mail, news, Web, and the like), routers, and modems attached to a permanent, high-speed Internet "backbone" connection. Subscribers can then dial into the local network to gain Internet access--without having to maintain servers, file for domain names, or learn Unix.

- **ISA**

Industry Standard Architecture Pronounced like Mr. Scott saying "Aye, sir" to Captain Kirk, ISA is the bus design that has been used in most PCs since IBM released the PC/AT more than a decade ago. It's a limited 8-bit and 16-bit bus, but it's so widely compatible that it has outlasted technologically superior--and much faster--bus standards like PCI. See also: PCI

- **IRQ**

interrupt request In PCs, an IRQ is a signal from a piece of hardware (such as a keyboard or sound card) indicating that it needs the CPU to do something. The interrupt request signals run along the IRQ lines to an interrupt controller that assigns priorities to incoming IRQs and delivers them to the CPU. Since the interrupt controller expects signals from only one device per IRQ line, if you have more than one device sending IRQ signals along the same line, you get an IRQ conflict that can freeze your machine. This is why assigning IRQs to new hardware as you install it is so important--and why it can be such a pain when it goes wrong. See also: Plug and Play

- **Plug and Play**

The Plug and Play (PnP) standards were developed by Microsoft, Intel, and other industry leaders to simplify the process of adding hardware to PCs. The standard's intention is to conceal unpleasant details, such as IRQ and DMA channels, from people who just want to add a new board, drive, or other chunk of hardware to their system. Some devices that don't seem that difficult to install also take advantage of PnP standards. A Plug and Play monitor, for example, can communicate with both Windows 95 and the graphics adapter to automatically set itself at the maximum refresh rate supported for a chosen resolution. PnP also ensures that the monitor will not be driven beyond its capabilities.

- **Java**

Sun Microsystems' Java is a programming language for adding animation and other action to Web sites. The small applications (called applets) that Java creates can play back on any graphical system that's Web-ready, but your Web browser has to be Java-capable for you to see it. According to Sun's description, Java is a "simple, object-oriented, distributed, interpreted, robust, secure, architecture-neutral, portable, high-performance, multithreaded, dynamic, buzzword-compliant, general-purpose programming language." And Sun should know.

- **JavaScript**

Designed by Sun Microsystems and Netscape as an easy-to-use adjunct to the Java programming language, JavaScript code can be added to standard HTML pages to create interactive documents. As a result, JavaScript has found considerable use in the creation of interactive Web-based forms. Most modern browsers, including those from Microsoft and Netscape, contain JavaScript support.

❖ **Java Virtual Machine**

The Java Virtual Machine (VM) is a program that interprets Java bytecodes into machine code. The VM is what makes Java portable--a vendor such as Microsoft or Sun writes a Java VM for their operating system, and any Java program can run on that VM.

❖ **JPEG**

Joint Photographic Experts Group This file format for color-rich images was developed by the Joint Photographic Experts Group committee. JPEG compresses graphics of photographic color depth better than competing file formats like GIF, and it retains a high degree of color fidelity. This makes JPEG files smaller and therefore quicker to download. You can choose how much to compress a JPEG file, but since it is a lossy format, the smaller you compress the file, the more color information will be lost. JPEG files can be viewed by a variety of downloadable software on both the PC and Mac. See also: GIF, lossless, lossy

❖ **JVM**

Java Virtual Machine A JVM acts as an interpreter between the Java bytecode and a computer's operating system. Using a JVM, you can run Java code on any number of different computer platforms, including Macintosh, Windows 95, and Unix. JVMs read and execute Java statements one at a time, however, so they are often slower than a just-in-time compiler.

❖ **kbps**

Your modem's speed is measured in the number of bits it can transfer in a second. Modems rated in kilobits per second are now the standard.

❖ **kilobyte**

Although kilo is Greek for a thousand, a kilobyte actually contains 1,024 bytes (that's 2 to the 10th power). In other words, a thousand bytes is actually less than a kilobyte. Remember that the next time you download a file. Abbreviated as K. See also: bit, byte, megabyte

❖ **laser printer**

While considerably more complex and expensive than most other common printer types, laser printers are capable of producing extremely high-quality text and graphics (including color) at fantastic speeds. At their most basic, laser printers apply an electrostatic charge to a drum inside the printer cartridge. A laser or a light-emitting diode then discharges portions of the drum to form the characters or graphics. Charged toner attaches itself to these discharged sections. A charged piece of paper is passed over the drum, transferring the toner. The toner is heated and fused to the sheet.

❖ **LAN**

local area network A local area network is a short-distance network used to link a group of computers together within a building. 10BaseT Ethernet is the most commonly used form of LAN. A piece of hardware called a hub serves as the common wiring point, enabling data to be sent from

one machine to another over the network. LANs are typically limited to distances of less than 500 meters and provide low-cost, high-bandwidth networking capabilities within a small geographical area. See also: Ethernet, 10BaseT, hub

- **LCD**

liquid-crystal display Created by sandwiching an electrically reactive substance between two electrodes, LCDs can be darkened or lightened by applying and removing current. Large numbers of LCDs grouped closely together can act as pixels in a flat-panel display.

- **leased line**

A dedicated private line that is used primarily to link two remote local area networks (LANs) together. Unlike frame relay, this line transmits data at only one speed depending on the purchased bandwidth. Customers pay a flat monthly rate for this service.

- **LED**

light-emitting diode In electronics terms, a diode is a semiconductor device through which current can go in only one direction. As a side effect, light-emitting diodes produce either visible or infrared light. LEDs require very little power and are often used as indicator lights, including (most likely) the drive access lights on your computer. LEDs are also found in some "laser-quality" printers.

- **Linux**

What do you do if you're not happy with any of the current operating systems? You build your own, of course. That's exactly what Linus Torvalds did. His freely distributed, Intel-processor-based alternative to Unix, Linux (pronounced lih-nucks), is currently used by hundreds of thousands--and possibly millions--of people around the world. While Linux began life primarily as a hobby for supergeeks, the operating system has made some inroads into corporate life, particularly as an inexpensive substitute for high-priced Unix Web servers. Linux is available from a number of vendors for several hardware platforms, including Intel x86, DEC Alpha, Sun Sparc, and Motorola PowerPC as well as IBM mainframes!

- **mbps**

megabits per second This abbreviation is used to describe data transmission speeds, such as the rate at which information travels over the Internet. Several factors can influence how quickly data travels, including modem speeds, bandwidth capacity, and Internet traffic levels.

- **megabyte**

Although mega is Greek for a million, a megabyte actually contains 1,048,576 bytes (1,024 x 1,024 bytes). In other words, a million bytes is actually less than a megabyte. Remember that the next time you buy a hard disk or try to fit files onto a floppy disk. (Although hard disk manufacturers usually calculate Megabyte as 1,000,000 bytes). Abbreviated as MB. See also: bit, byte, kilobyte

- **memory bandwidth**

Generally, bandwidth refers to data-carrying capacity and is expressed in cycles per second or Hertz (Hz). In the case of RAM, bandwidth is a function of its rated speed and the size of its data path. See also: bandwidth, RAM

- **MFLOPS**

mega-floating point operations per second Used as a measure of how powerful your computer is, MFLOPS gauge the capability of your system to deal with floating-point math instead of raw instructions.

- **MHz**

megahertz A megahertz is 1 million complete cycles per second. This unit is most commomly used to measure transmission speeds of electronic devices, such as the clock speed of a microprocessor, the small computer chip that handles data-related tasks.

- **microprocessor**

The microprocessor handles the logic operations in a computer, such as adding, subtracting, and copying. A set of instructions in the chip design tells the microprocessor what to do, but different applications can give instructions to the microprocessor as well. Chip speeds are measured in megahertz (MHz), so a 120-MHz chip is twice as fast as a 60-MHz chip. However, that doesn't mean your computer will run all tasks twice as fast, as speed is also influenced by other factors, such as the design of the software you're running, the operating system you're using, and so on. The first microprocessor, the 8080, was created by Intel. Other early microprocessors included Motorola's 6800 and Rockwell's 6502. The most popular microprocessors today are Motorola's PowerPC and Intel's Pentium. See also: megahertz

- **middleware**

This software manages the communication between a client program and a database. For example, a Web server connected to a database can be considered middleware--the Web server sits between the client program (a Web browser) and a database. The middleware allows the database to be changed without necessarily affecting the client, and vice versa.

- **millisecond**

A millisecond is one-thousandth of a second. The term is most commonly used to measure data access speeds, such as the amount of time it takes to retrieve data from a hard disk, a CD-ROM drive, or a floppy drive. Milliseconds are sometimes confused with mbps (megabits per second), which describe data transmission speeds. See also: mbps

- **MIME**

multipurpose Internet mail extensions MIME has absolutely nothing to do with Marcel Marceau. It's a way to extend the power of Web browsers to handle graphics, sound, multimedia--anything but text. (Remember, HTML handles nothing but text--everything else is an extension.) MIME is also used for binary email attachments. Browsers recognize MIME types in categories and file

types, separated by a slash (such as image/gif). If you've registered a MIME type, the browser decodes the file and launches a helper application.

- **MIPS**

millions of instructions per second Used as a measure of how powerful your computer is, MIPS gauge the capability of your system to handle instructions. See also: MFLOPS

- **mirror server**

Sometimes a Web server will receive more traffic than it can handle. When this happens, the server's administrator may add extra servers--containing identical data--to accommodate the flow. These duplicates are called mirror servers. By adding mirror servers and telling users how to access them, an administrator can keep users from receiving error messages or unacceptably slow response times when they try to access a site. Mirror servers also act as backups if the primary site goes down. See also: server

- **mirror site**

Because the Internet population has exploded in recent years, a lot of archive servers can't cope with the load. One solution is to create an exact copy of a server--a process called mirroring. Mirror sites divert some of the traffic from the original site. It's not unusual to find a dozen or more mirrors of busy ftp sites. See also: ftp

- **Moore's law**

More than 25 years ago, when Intel was developing the first microprocessor, company cofounder Gordon Moore predicted that the number of transistors on a microprocessor would double approximately every 18 months. To date, Moore's law has proven remarkably accurate.

- **motherboard**

Every system has one: The motherboard is the largest printed circuit board in your computer. It generally houses the CPU chip, the controller circuitry, the bus, and sockets for additional boards, which are called daughterboards. If you have a horizontal-style computer, the motherboard is generally the one at the bottom of the computer's box. If you have a tower-configuration box, it's along one of the vertical sides.

- **MP3**

MPEG-1, Layer 3 MP3 is a codec that compresses standard audio tracks into much smaller sizes without significantly compromising sound quality.

- **MPEG**

Moving Pictures Experts Group MPEG is a standard for compressing sound and movie files into an attractive format for downloading--or even streaming--across the Internet. The MPEG-1 standard streams video and sound data at 150 kilobytes per second--the same rate as a single-speed CD-ROM drive--which it manages by taking key frames of video and filling only the areas that change between the frames. Unfortunately, MPEG-1 produces only adequate quality video, far

below that of standard TV. MPEG-2 compression improves things dramatically. With MPEG-2, a properly compressed video can be shown at near-laserdisc clarity with a CD-quality stereo soundtrack. For that reason, modern video delivery mediums, such as digital satellite services and DVD, use MPEG-2.

- **NIC**

network interface card An adapter card that physically connects a computer to a network cable.

- **NNTP**

network news transfer protocol Usenet news articles can't be just posted and accessed willy-nilly, so they conform to this protocol that runs interference between newsreaders and news servers. NNTP dictates the way in which news articles are distributed, queried, retrieved, and posted.

- **null-modem cable**

A special type of computer cable that lets you hook up two computers to communicate via their serial ports. It's called a "null-modem" cable because it eliminates using modems and phone lines for hooking together nearby computers. Null-modem connections are useful for file transfers, remote control, or special applications like the dual-screen mode of Microsoft's PowerPoint 97.

- **object-oriented**

A method of software-development that groups related functions and data into reusable chunks. Properly handled, object-oriented programming can reduce development time on new projects.

- **OCR**

optical character recognition When your computer gets a fax or scans in text, all it sees are graphical bits on a virtual page. That text is not usable, searchable, or editable. If you pass the page through an OCR program, the software converts the shapes on it into a text document. However, few documents are perfectly recognized and the errors are frequent if the type is small or the scan unclear. But the conversion is often faster than typing text manually.

- **OS**

operating system A computer by itself is essentially dumb bits of wire and silicon. An operating system knows how to talk to this hardware and can manage a computer's functions, such as allocating memory, scheduling tasks, accessing disk drives, and supplying a user interface. Without an operating system, software developers would have to write programs that directly accessed hardware--essentially reinventing the wheel with every new program. With an operating system, such as Windows NT or Mac OS 8, developers can write to a common set of programming interfaces called APIs and let the operating system do the dirty work of talking to the hardware.

- **optical fiber**

Optical fiber cables consist of thin filaments of glass (or other transparent materials), which can carry beams of light. A laser transmitter encodes frequency signals into pulses of light and sends them down the optical fiber to a receiver, which translates the light signals back into frequencies.

Less susceptible to noise and interference than other kinds of cables, optical fibers can transmit data greater distances without amplification. But because the glass filaments are fragile, optical fiber must be run underground rather than overhead on telephone poles

- **packet**

While it may seem as though you send or receive a continuous stream of data every time you use the Internet, you don't. Instead, data is chopped up into pieces called packets. These packets contain information about which computer sent the data and where the data is going. If a packet runs into a problem during its trip, it can attempt to find another route. When all the packets get where they're going, the recipient computer puts them together again.

- **parallel port**

Look on the back of any PC and you'll see a 25-pin RS-232C connector, better known as a parallel port. Theoretically eight times faster than serial ports, parallel ports are also called the printer ports, since that's what you usually find attached to them (although you can get tape backup units, CD-ROM drives, scanners, and other devices in parallel port format these days). Parallel ports come in several configurations. For example, 4-bit ports output 8 bits of data at a time but receive only 4 bits, while 8-bit ports handle 8 bits of data both ways. "Pseudo 8-bit ports" work at similar speeds, but only with more sophisticated software and peripherals.

- **Phase Alternate Line (PAL)**

The television broadcast standard throughout Europe (except in France, where SECAM is the standard). This standard broadcasts 625 lines of resolution, nearly 20 percent more than the U.S. standard, NTSC, of 525.

- **PCI**

Peripheral Component Interconnect If you have a Pentium system, it's extremely likely that it runs a self-configuring PC local bus called PCI. Designed by Intel, PCI has gained wide acceptance (even by Apple, in its PowerPC series). It beats out the VESA Local Bus spec from a technical standpoint and will presumably win out in the long run. The bottom line: if you have a Pentium, make sure any add-in board you buy is a PCI device.

- **Perl**

practical extraction and report language The programming language of choice for writing Web server applications, Perl is used for creating interactive forms and a slew of other CGI programs. This free-licensed language comes in versions for Windows NT, Novell NetWare, and Unix. Perl scripts are available free of charge all over the Internet. See also: CGI

- **peer-to-peer network**

A network where there is no dedicated server. Every computer can share files and peripherals with all other computers on the network, given that all are granted access privileges. Such a network is practical only for small workgroups of less than a dozen computers.

❖ **Ping**

packet Internet groper When submarine crews wanted to test the distance of an object, they would send a sonar "ping" and wait to hear the echo. In the computer world, Ping is a program that "bounces" a request off of another computer over a network to see if the remote computer is still responding. If the ping comes back, the remote computer is still alive. See also: TCP/IP

❖ **pixel**

picture element The image displayed on monitors or in a graphic produced by a scanner or paint program is made up lots of dots called pixels. Collectively, the number of pixels displayed is referred to as the image's resolution. A pixel on a monitor is a number of red, green, and blue phosphor dots. These dots are "excited" to varying degrees by the monitor's three electron guns, and the results mix additively to generate a specific color. By manipulating large numbers of pixels in precise ways, patterns emerge to make up an identifiable picture.

❖ **POP**

Post Office Protocol The current champ in Internet email mailbox access standards, but its limitations--basically, you connect to a server and download all your messages, which are then deleted from the server--discourage flexibility. Of course, some clients let you leave all messages on the server, and/or refuse to download messages above a certain size. Still, as messages become longer--with multimedia (such as sound or video) objects and the likes--we'll want some flexibility in what we retrieve and when we retrieve it. That's where IMAP comes in. The current version of POP is POP3.

❖ **PPP**

point-to-point protocol PPP is the Internet standard for serial communications. Newer and better than its predecessor, SLIP, PPP defines how your modem connection exchanges data packets with other systems on the Internet. See also: SLIP

❖ **SLIP**

serial line Internet protocol SLIP is a standard for connecting to the Internet with a modem over a phone line. It has serious trouble with noisy dial-up lines and other error-prone connections, so look to higher-level protocols like PPP for error correction. See also: PPP

❖ **PRI**

Primary Rate Interface This ISDN service is used mainly by Internet service providers (ISPs) and businesses. Why? Because it provides a lot of bandwidth: 23 B channels with 64-kbps throughput capability and one 64-kbps D channel for call signaling and setup. It was designed for transmission through a standard North American T1 trunk and has throughput capability to 1.472 mbps. If you see the notation 23B+D, that's another way of saying the same thing.

❖ **protocol**

Computers can't just throw data at each other any old way. Because so many different types of computers and operating systems connect via modems or other connections, they have to follow communications rules called protocols. The Internet is a very heterogenous collection of networked computers and is full of different protocols, including PPP, TCP/IP, SLIP, and ftp. See also: ftp, PPP, SLIP, TCP/IP

- **proxy servers**

A proxy server is a system that caches items from other servers to speed up access. On the Web, a proxy first attempts to find data locally, and if it's not there, fetches it from the remote server where the data resides permanently. See also: server

- **public domain**

Of all the kinds of software or information you can download, public domain has the fewest strings attached. With shareware, you're expected to pay a fee. With freeware, you may face other restrictions, and there's still a copyright attached. With public domain downloads (also called downloads in the pubic domain), there are no copyright restrictions whatsoever. See also: shareware, freeware

- **BRI**

Basic Rate Interface This ISDN service is currently offered by many telephone companies for home users. With only slight modifications to existing telephone service, ISDN BRI allows up to 128-kbps throughput. It consists of two 64-kbps B channels for carrying data or voice and one 16-kbps D channel for call setup and signaling. BRI service is therefore sometimes referred to as 2B+D. See also: ISDN

- **QuickTime**

Developed by Apple Computer, QuickTime is a method of storing sound, graphics, and movie files. If you see a MOV file on the Web or on a CD-ROM, you'll know it's a QuickTime file. Although QuickTime was originally developed for the Macintosh, player software is now available for Windows and other platforms. If you don't have a QuickTime player, you can always download versions for either Mac or PC from Apple's Web site.

- **RAID**

redundant array of independent (or inexpensive) disks A RAID provides convenient, low-cost, and highly reliable storage by saving data on more than one disk simultaneously. At its simplest, a RAID-1 array consists of two drives that store identical information. If one drive goes down, the other continues to work, resulting in no downtime for users. RAID-1 isn't a very efficient way to store data, however. To save disk space, RAID-3, -4, and -5 "stripe" data and parity information across multiple drives (RAID-3 and -4 store all parity data on a single drive). If a single disk fails, the parity information can be used to rebuild the lost data. Unfortunately, there is a performance trade-off: depending on the RAID type used, a RAID will be slower than a single drive at either reading or writing data.

❖ **RAM**

random access memory When you run an application like Microsoft Word, the program is called up from its permanent storage area (like the hard drive, floppy disk, or CD-ROM) and moved into the RAM, where it sends requests to the CPU. Using the faster PC100 memory preferred by 350-, 400-, and 450-MHz Pentium II processors means your information spends less time in line before being processed. (PC100 chips are rated to perform at bus speeds up to 100 MHz.) Your computer should have as much RAM as you can afford so it can work efficiently. It also pays to have lots of memory in your system because some operating systems, including Windows 95 and 98, swap applications from memory to your hard drive when the RAM gets filled. That means that instead of having your speedy RAM sending out requests, the OS sends the work to be done by the much slower hard drive. See also: DRAM, EDO RAM, SRAM, virtual memory, von Neumann architecture

❖ **SRAM**

static RAM SRAM is like DRAM on steroids. Since it stores its data in capacitors that don't require constant recharging to retain their data, it performs better (though it is pricier to produce). SRAM is typically used for secondary caches because its speed falls somewhere between that of DRAMs and a CPU-based cache (typically 8ns to 20ns, as opposed to 60ns to 80ns for DRAM). See also: EDO RAM, DRAM, RAM

❖ **virtual memory**

What do you do when you run out of real random access memory (RAM)? Easy. Pass it off to virtual memory. To do this you need a virtual memory manager (usually a function of the operating system) that maps chunks of data and code to storage areas that aren't RAM. Virtual memory is really a part of your hard disk called a swap file, dedicated as a storage area for bits of data in RAM that aren't being used much. By freeing up RAM, you're virtually increasing the amount of working memory available to you. See also: RAM, swap file

❖ **von Neumann architecture**

Most computers use the stored-program concept designed by Hungarian mathematician John von Neumann. In it, you store programs and data in a slow-to-access storage medium (such as a hard disk) and work on them in a fast-access, volatile storage medium (RAM). This concept, however, has an attendant bottleneck: it's designed to process instructions one after the other instead of using faster parallel processing. See also: RAM

❖ **Registry**

The Windows Registry stores system configuration details so that Windows looks and behaves just as you want it to. The Registry stores user profile information such as wallpaper, color schemes, and desktop arrangements in a file called user.dat. And it stores hardware- and software-specific details, such as device management and file extension associations, in a file called system.dat. In many ways, the Registry (Windows 95 and later) replaces functions of win.ini and system.ini from earlier versions of Windows, though these files persist because so many Windows

applications refer to them. Registry details can be edited using a program called RegEdit (which ships with Windows 95 and later) and exported to text format as a file with the extension REG.

- **resolution**

Resolution is a measure of graphics that's used to describe what a printer can print, a scanner can scan, and a monitor can display. In printers and scanners, resolution is measured in dots per inch (dpi)--the number of pixels a device can fit in an inch of space. A monitor's resolution refers to the number of pixels in the whole image, because the number of dots per inch varies depending on the screen's dimensions. For example, a resolution of 1,280 by 1,024 means that 1,024 lines are drawn from the top to the bottom of the screen, and each of these lines is made up of 1,280 separate pixels--and in turn, each dot may have any number of combinations of red, green, and blue intensities. Common resolutions in the PC world include 640 by 480 (also called VGA resolution; appropriate for a 14-inch monitor), 800 by 600 (appropriate for a 15-inch monitor), 1,024 by 768 (appropriate for a 17-inch monitor), and 1,280 by 1,024. See also: dpi, pixel

- **RGB**

RGB refers to the so-called scientific hues--the additive primary colors red, green, and blue--that, when mixed together in equal amounts, create white light. Television sets and computer monitors display their pixels based on values of red, green, and blue.

- **RJ-11**

This is the standard telephone connector--a tab snaps into the socket and has to be pressed to remove the connector from the wall. An ordinary phone circuit uses two wires. The RJ-11 jack has room for up to four wires, but at a glance it's easy to mistake with the larger RJ-45 jack, which can house up to eight wires. See also: RJ-45

- **RJ-45**

RJ-45 connectors look a bit like a standard phone connector but are twice as wide (with eight wires). RJ-45s are used for hooking up computers to local area networks (LANs) or for phones with lots of lines. See also: LAN, RJ-11

- **ROM**

read-only memory ROM is a storage chip that typically contains hardwired instructions for use when a computer starts (boots up). The instructions--contained in a small program called the BIOS (basic input/output system)--load from ROM and start up the hard disk so that the operating system (OS) can be loaded and the whole shooting match can begin. Some ROM chips can be updated with new BIOS instructions--but unless you hear them called EEPROMs or flash BIOSs, the likelihood is, they can't be. See also: BIOS, EEPROM

- **router**

This piece of hardware does what it says: it routes data from a local area network (LAN) to a phone line's long distance line. Routers also act as traffic cops, allowing only authorized machines to transmit data into the local network so that private information can remain secure. In addition to

supporting these dial-in and leased connections, routers also handle errors, keep network usage statistics, and handle security issues. See also: LAN

- **RS-232**

Recommended Standard 232 This was originally a nine-wire interface standard for teletype machines from the Electronics Industry Association. Now in its third revision (RS-232-C), it's the standard for computer serial-port transfers. The RS-232 standard is probably the only computer component that's 40 years old and still working. One wire is used as the ground; the rest are dedicated to detecting carrier signals, managing the timing of data transfer, oh, and sending and receiving data.

- **SCSI**

Small Computer System Interface While the PC was settling for rankly inferior alternatives, the Mac adopted SCSI as its expansion standard. With SCSI, you can add up to seven new devices to your computer and depend on them to deal with single-interface issues by themselves. It's a robust standard, and it's rendered even more so with its latest implementation, SCSI-2. But it requires some system overhead, slows down your computer's start-up, and demands that during installation you handle device ID administration and a process called termination that closes the SCSI circuit. Pronounced "scuzzy" by those in the know.

- **SDRAM**

synchronous dynamic RAM Sending data from main memory to the system processor is consistently one of the biggest performance bottlenecks in any PC. Even the fastest standard DRAM and EDO memory cannot keep up with the 66-MHz bus speeds used on many Pentium systems. SDRAM incorporates new features that allow it to keep pace with bus speeds as high as 100 MHz. It does this primarily by allowing two sets of memory addresses to be opened simultaneously. Data can then be retrieved alternately from each set, eliminating the delays that normally occur when one bank of addresses must be shut down and another prepared for reading during each request. See also: DRAM, EDO RAM, RAM, SRAM, SGRAM, WRAM

- **SGRAM**

synchronous graphics RAM SGRAM contains the speed-enhancing features of SDRAM and adds graphics capabilities that enhance 3D graphics performance. Like SDRAM, SGRAM can work in sync with system bus speeds up to 100 MHz. See also: DRAM, EDO RAM, RAM, SRAM, SDRAM, WRAM

- **WRAM**

window RAM No, it has nothing to do with Microsoft's operating environment. Instead, WRAM is a variation on dual-ported memory technology that includes a larger bandwidth and more graphics-handling features than VRAM has, making WRAM useful for graphics applications. See also: DRAM, EDO RAM, RAM, SRAM, VRAM

- **VRAM**

video RAM This type of RAM sits on the better class of graphics display adapters. Unlike its general-purpose cousin dynamic RAM (DRAM), VRAM has dual ports--a design that can read and write data at the same time and is thus faster than DRAM. See also: DRAM, RAM

- **search engine**

When a user enters text into a search form, a program called a search engine analyzes the text and searches for matching terms in an index file, which was created using a search indexer. The search engine returns the results of its search using a results listing.

- **search form**

An HTML page that lets users type in search terms and set various search options. For example, the main page on Snap contains a search form. When a user enters a search, a program called a search engine compares the search criteria against an index file, which was created using a search indexer. The search engine returns the results of its search using a results listing.

- **semiconductor**

A class of materials that allow electrical current to flow through them under certain conditions. Semiconductors are used to create common electronic components, such as diodes and transistors. See also: diode, transistor

- **transistor**

One of the most important inventions in history, the transistor can be found in nearly every common electronic device manufactured today--radios, TV sets, cellular phones, computers, and so on. Originally created in the late 1940s by Bell Labs, the transistor was hailed as a smaller, less-expensive, and cooler-running replacement for the vacuum tubes then commonly used to amplify current in electronic devices. Today, millions of transistors are often packed into silicon chips to create the processors used in modern computers.

- **shareware**

Shareware is the wonderful alternative to commercial software. Available from centralized archives on the Internet and local bulletin board systems (or sometimes via CD-ROM or floppy), shareware is copyrighted but works on the honor system. You have a specified time period to try out the software for free; if you continue to use it, you're expected to register the program and pay a fee to its developer. (Some programs are partially disabled, stop working after a set period of time, or contain "nag screens" that pop up frequently to encourage you to register.) Registration fees are usually no more than $50, and some selfless developers ask only that you send a postcard letting them know you like their product. Registering often gets you full documentation or free software updates--not to mention a clear conscience. Shareware that doesn't involve a fee is called freeware. See also: freeware, public domain

- **SHTTP**

secure hypertext transfer protocol This protocol was developed by Enterprise Integration Technologies to keep your moolah safe on its way from your wallet to a commercial transaction on the Internet.

- **SIMM**

single in-line memory module SIMMs are the most widespread form of RAM available. They're about 10cm by 2cm, and they sit in rows at either 90 degrees or 45 degrees to your motherboard. You can generally get them in 1MB to 32MB configurations.

- **S/MIME**

Secure Multipurpose Internet Mail Extensions S/MIME defines a means to make email messages more secure by adding both digital signatures and encryption. Using S/MIME-compliant email packages, users can make sure that a message in fact comes from the supposed sender (thanks to the signature), and that no one else could read the message before it arrived (thanks to encryption).

- **SMTP**

simple mail transfer protocol When you're exchanging electronic mail on the Internet, SMTP is what keeps the process orderly. It's a protocol that regulates what goes on between the mail servers. See also: IP, NNTP, TCP/IP

- **snail mail**

This term is used by supercilious fans of email to describe the regular paper-based mail service. Since the delay between sending email and receiving it can be as little as a few seconds, regular mail seems a lot slower by comparison. See also: email

- **SQL**

Structured Query Language A type of programming language used to construct database queries and perform updates and other maintenance of relational databases, SQL is not a full-fledged language that can create standalone applications--but it is strong enough to create interactive routines in other database programs. If you're looking into buying relational database software, make sure it has SQL support. Incidentally, SQL was developed as a result of an IBM project called Structured English Query Language, so to this day it is pronounced "sequel," not "squeal."

- **swap file**

A swap file is an area on your hard disk used as virtual memory. It's called a swap file because virtual memory management software swaps data between it and main memory (RAM). See also: virtual memory, RAM

- **T1**

If ISDN isn't enough digital carrier for you, T1 offers faster speeds. T1 is a term coined by AT&T for a system that transfers digital signals at 1.544 megabits per second (as opposed to ISDN's mere 64 kilobits per second). Of course, if T1 doesn't cut it, there's always T3. (T2 seems to have been bypassed altogether.) See also: ISDN, POTS, T3

❖ **T3**

When you're transferring data across a digital carrier, T3 is the premium way to go. It's not just three times the capacity of T1, as the name suggests--it's almost 30 times the capacity. It can handle 44.736 megabits of digital data per second. See also: ISDN, T1

❖ **TCP/IP**

transmission control protocol/Internet protocol These two protocols were developed by the U.S. military to allow computers to talk to each other over long distance networks. IP is responsible for moving packets of data between nodes. TCP is responsible for verifying delivery from client to server. TCP/IP forms the basis of the Internet, and is built into every common modern operating system (including all flavors of Unix, the Mac OS, and the latest versions of Windows). See also: protocol, data packet, client, server

❖ **Turing experiment**

Named for computing pioneer Alan Turing, this is one element of testing the robustness of artificial intelligence; it's also called a Turing test. Back in the 1950s, Turing believed that by the end of this century a computer with the right program could engage in a written conversation, and for 5 minutes pass for a real human about 70 percent of the time. This test (now held annually for the Loebner Prize) involves participating in a written conversation via email or online chat. The unseen correspondent may be a person or a program, and if the human participant believes it's another person when it's really a program, the software is classified as possessing true artificial intelligence.

❖ **TWAIN**

While there are some who claim that TWAIN stands for toolkit without an interesting name, in fact it stands for nothing but itself. But what is it? TWAIN is an interface standard that should be on the checklist of anyone buying a scanner or OCR, graphics, or fax software. If your scanner supports TWAIN, you can use any TWAIN-compliant software to run it. Like a corporate raider, TWAIN's signature command is Acquire--if you spot the Acquire option under a program's File menu, you know the software is TWAIN-compliant. See also: OCR

❖ **Unix**

Described by one of its developers as "a weak pun on Multics" (which was an experimental, time-sharing operating system at Bell Labs in the 1960s), Unix took off in the early 1970s as a general-purpose operating system. Since much of the Internet is hosted on Unix machines, the OS took on a new surge of popularity in the early 1990s. Unix comes in many flavors--including Xenix, Ultrix, GNU, and Linux--and runs on a variety of platforms, which makes its development a subject of widespread discussion. But the truly great debate involves how to style the word itself: should it have an initial capital (Unix)? Or should it be in all caps (UNIX)? Since the operating system itself is case-sensitive, the debate rages. Bell Labs' implementation of Unix is trademarked in all caps; for the other implementations, it's optional.

❖ **URL**

uniform resource locator universal resource locator URLs are the Internet equivalent of addresses. How do they work? Like other types of addresses, they move from the general to the specific (from zip code to recipient, so to speak). Take this URL, for example: http://www.vu.edu.pk/admission/admission.htm First you have the protocol: http:/ then the server address or domain: /www.vu.edu.pk and finally the directory: /admission/ in which the file admission.htm resides. Two debates rage: first, does the U stand for uniform or universal? Universal was the original definition of choice but was deemed by most to be too ambitious, and the more frequently used uniform was instated by the now-defunct URI Working Group. Second, is URL pronounced "you are ell," or does it rhyme with hurl? Both pronunciations are widely used.

- **VBScript**

Visual Basic Scripting Edition (VBScript) is a programming language developed by Microsoft for creating scripts (miniprograms) that can be embedded in HTML Web pages for viewing with Internet Explorer. These scripts can make Web pages more interactive. VBScript also works with Microsoft ActiveX Controls, allowing Web site developers to create forms, interactive multimedia, games, and other Web-based programs. VBScript is similar in functionality to JavaScript and is a subset of the widely used Microsoft Visual Basic programming language. See also: ActiveX, JavaScript, Visual Basic Click Here

- **VPN**

Virtual Private Network A Virtual Private Network, or VPN, is a private network of computers that's at least partially connected by public phone lines. A good example would be a private office LAN that allows users to log in remotely over the Internet (an open, public system). VPNs use encryption and secure protocols like PPTP to ensure that data transmissions are not intercepted by unauthorized parties.

- **WAN**

wide area network Take two local area networks, hook them together, and you've got a WAN. Wide area networks can be made up of interconnected smaller networks spread throughout a building, a state, or the entire globe. See also: LAN

- **WAV**

Pronounced "wave," this is the Windows standard for waveform sound files. WAV files predictably have the extension .wav.

- **World Wide Web**

Also known as the WWW, the W3, or most often simply as the Web, it originally developed by CERN labs in Geneva, Switzerland. Continuing development of the Web is overseen by the World Wide Web Consortium. The Web can be described (dryly) as a client/server hypertext system for retrieving information across the Internet. On the Web, everything is represented as hypertext (in HTML format) and is linked to other documents by their URLs. The Web encompasses its native http protocol, as well as ftp, Gopher, and Telnet. The best way to learn about it, however, is to try it for yourself.

❖ **WYSIWYG**

What You See Is What You Get A catchphrase from the old TV show Rowan and Martin's Laugh-In that became a desktop publishing byword, WYSIWYG (pronounced "whizzy-wig") refers to any technology that enables you to see images onscreen exactly as they will appear when printed out. As scalable screen and printer fonts have become more sophisticated, and as graphical user interfaces have improved their display, people have come to expect everything to be WYSIWYG. But it isn't always the case--and certainly wasn't in the 1980s, when this term was first applied.

❖ **XML**

Extensible Markup Language XML is the Extensible Markup Language, a system for defining specialized markup languages that are used to transmit formatted data. XML is conceptually related to HTML, but XML is not itself a markup language. Rather it's a metalanguage, a language used to create other specialized languages.

❖ **ZIF socket**

zero insertion force socket Here's the problem: when you're swapping microprocessor chips in and out of their sockets (as you do when you're upgrading your CPU), you're in danger of bending a pin or a whole row of pins with the pressure you exert. When the chip is a Pentium worth hundreds of dollars, this is extremely bad news. So the industry introduced ZIF sockets, which use leverage instead of brute force to seat and unseat chips--thus taking the pressure out of chip relocation.

❖ **APM**

Automated Power Management. Series of techniques used to reduce the power consumption, especially on portable computers, in order to lengthen their autonomy.

❖ **Autentication Service**

A mechanism, analogous to the use of passwords on time-sharing systems, for the secure authentication of the identity of network by servers, and vice versa, without presuming the operating system integrity of either (e.g., Massachusetts Institute of Technology's Kerberos) .

❖ **ATM**

Asynchronous Transfer Mode. A wide-area network technology ; a transfer mode for switching and transmission that efficiently and flexibly organises information into cells. It is asynchronous in the sense that the recurrence of cells depends on the required or instantaneous bit rate. Thus, empty cells do not go unutilised when data is waiting. ATM's powerful flexibility lies in its ability to provide a high-capacity, low-latency switching fabric---for all types of information, including data, video, image and voice --- that is protocol , speed and distance independent. Of the plethora of benefits ATM will accrue to its users, probably the most significant is that it represents an enduring, architectural approach due to its inherent scalability. It scales well from small to large systems, from very low to very high transmission speeds (more than 100 Mbps), and from local-area, through metropolitan-area, to wide-area networking environments. This promising technology is not anticipated to be widely used until 1998.

❖ **80x8x**

A family of Intel microprocessors used in IBM and IBM-compatible PCs and workstations.

❖ **Absolute URL**

The full Internet address of a page or other World Wide Web resource. The absolute URL includes a protocol, such as "http," a Network location, an optional path and a file name, for example: http://anything.youwant.com

❖ **Address**

For once, it's not your home address but the unique identifier assigned to a Web page. Those who want to really show off like to refer to it as their URL - Sorry, no relation to an Earl.

❖ **Analogue Signal**

An electrical signal that varies according to the characteristics of the information it presents. The standard telephone line in your house transmits an analogue signal. It's the job of your modem to convert the digital computer signal into an analogue one so that it can be sent down your telephone line. Clever stuff!

❖ **Apple**

They used to say that if you eat one a day it'll keep the doctor away. Well at one time the famous company that makes the Apple Mac range of computers used to sell one a day. Unfortunately, they're not IBM compatible, even though they do have a limited ability to read IBM disks. Graphic Designers, Art Students, Publishers and the Media people like them a lot, but the majority of us Brits still prefer a more user friendly PC or Laptop.

❖ **Application**

What you fill in when you apply for a job. Alternatively, a posh name for a program used by you on your computer to create something. Examples could be your word processor, painting program, desktop publisher or diary. The list is endless. All modern computers come with several applications pre-installed. If you want more, you'll have to dip into your wallet!

❖ **Archive**

Archives are usually large files that have been compressed, and sometimes they may even contain several other smaller files. Some examples of archive file extensions are: zip, tar, arc, lzh and ARJ. 'Winzip' is the most popular computer program for home users for compressing files and it's often used to reduce the size of large ones prior to them being attached to E-Mails. A File that has been archived in this way will have a .zip file extension after it's name, such as: areyou.zip withit.zip ok.zip

❖ **ASP**

A document that contains embedded server side scripting (known as an active server page). On the client side (i.e. your computer) an ASP is a standard html web document that can be viewed in any Web Browser on any computer.

- **ATAPI**

Advanced Technology Attachment Packet Interface: What a mouthful! It's the recognized standard for connecting a CD-ROM Drive to an enhanced IDE adapter (which is what most computers use to handle their hard disk/s) and now that ATAPI is accepted as the standard it's made the installation of a CD-Rom drive a much easier task.

- **Backup**

It's a second copy of an important file, letter or data etc, which can be used to restore your computer or application after a crash, etc.

- **Beta**

An advanced (pre-release) version of a new software package that is still being developed. A Beta version is often supplied to a selected group of testers for evaluation, criticism and bug testing. Their comments and observations are analyzed by the software manufacturers, changes are sometimes made then it is released to the general public.

- **Body**

In E-mail terms it means the part of the message containing the text content. The term Body is also used in Web page design where it refers to the html code hidden within the <body> tags of a Web Page. To see a demonstration of this, Right-Click this page, select View Source, Click on Yes (to enable Notepad) and you'll see what I mean.

- **Bookmark**

Virtual bookmarks work pretty much the same as the real ones you use in your favorite novel. They record a URL or Web page to allow you to refer back to it at a later date. Bookmarks are also used to link one area (or subject) on a Web page to another one (usually on the same page) by using a hypertext link (funny little hand). We often use Bookmarks to link the Title Headings on our pages to sections of relevant text within that page.

- **Boot Diskette**

A floppy disk that has the special system files on it that are needed to start up the computer using its own operating system. Since CD Writers have became popular, it's also possible to make a Boot CD-Rom which will do the same thing.

- **Check-box**

We've all seen these little square boxes that once clicked into, display a tick or a cross to indicate that the item has been selected. An essential item when filling in a questionnaire or when adding things to your trolley when online shopping.

❖ **Chipset**

A group of computer 'chips' working together to perform specific functions such as 3D graphics.

❖ **Client**

In a Client Server relationship, the client is the computer running programs or applications from the server, or accessing files from it, and does the bulk of the processing of this information.

❖ **Clipart**

These are ready made images, illustrations or pictures which are usually supplied on a CD-ROM. Sometimes they are supplied free with new software packages. Occasionally you may wish to install them on your PC, but more often than not (due to the shear numbers of images on these types of CD's) it's better to just import the ones you want as and when you need them. Clipart is usually supplied 'Royalty Free' and is free of copyright.

❖ **Clusters**

The little areas on the computer's hard drive were files are stored and organized into sectors and blocks

❖ **Compress**

Not the cold type you put on a swelling, but the act of discarding redundant or semi redundant information from a file in order to reduce its size. For example a picture may be compressed by anything up to 40% but a text document will compress down to 80% of its original size. Large pictures to be attached to E-Mails need compressing.

❖ **Computer**

In Theory: A sophisticated electrical machine that can be programmed to manipulate symbols and perform complex and repetitive tasks efficiently, repeatedly, precisely, and reliably. It must also have the capability to store data, access it when requested and display it as it was - when it was originally saved by the user. In Practice, does yours do all of this? If it doesn't, it's time to think about updating it!

❖ **Algorithm**

A step-by-step method of accomplishing a task. For instance, a sorting algorithm may be like this: 1) Progress down your list from the top; 2) When you encounter something that is less than the last item swap the two; 3) Repeat Steps 1 and 2 until you reach the end of your list. This is commonly known as a Bubble Sort, as it seems to bubble items to the top of the list.

❖ **CC (carbon copy)**

A method of sending a copy of an e-mail to someone, but implying that the person is not the direct recipient. For example, you send an e-mail with instructions to a group you manage, and CC it to your boss so that he or she knows what's going on but understands that the instructions in the mail were not meant for him or her to carry out. When you carbon copy someone in an e-mail, the

recipients in the "To" field of the e-mail are aware of the names in the CC field. If you want to keep names secret from the To and CC recipients, you would use "BCC," or blind carbon copy.

- **FAT32; FAT32 file system:**

FAT 32 uses the hard drive more efficiently and organizes the file space into smaller sections that can give you 10% or 15% more room on your hard disk drive. Additionally, since the hard drive sections are smaller, this produces a boost in the operating system speed. FAT32 supports up to a two terabyte hard drive, limited only by your BIOS interface. FAT32 keeps a better record of the hard drive files and is more reliable than FAT.

- **File**

Not the type you might use to take the sharp edges off a piece of metal or wood. This one's a collection of computer data stored all together in a specific place on your hard drive (or floppy disk etc) until it's required again. Examples could be a text document that you created in MS Word, or a complete installed program or application.

- **Folder**

Similar to its paper equivalent, it's a place where files are stored on the computers hard drive.

- **GIF**

Graphics Interchange Format: This has become the standard file format to be used when displaying fixed or animated images on an Internet Web Page. It usually contains up to 256 colours and this allows for small accurate images to be statically displayed or in the case of animated images, several moving ones shown continuously. I'm sure you've seen lots of these. The gif file was originally defined by CompuServe.

- **Hard Coded**

These are computer chips or memory that contain software instructions which have been programmed into them when they were manufactured on the production line.

- **Hard Disk (or Hard Drive)**

The main storage area on the computer where data is stored. Usually referred to as C: drive. On early Windows 95 PC's the size of the hard disk used to be measured in Megabytes but since the addition of a CD-Rom (or DVD) as standard equipment and the increase in performance provided by faster Pentium Processors and other internal improvements, hard disk capacity has dramatically increased. A good 'off-the-shelf' computer now comes with at least 10 GB as standard.

- **Hard Wired**

This refers to devices, ports and chips etc, all connected to the motherboard of the computer as it's being assembled. It doesn't include any cards that may be plugged in and out after production is completed in the factory.

- **Header**

What the goalie has to fear from a good Centre Forward. In E-mail terms, it's the part of the message indicating who the sender is, and usually included some other brief details such as the subject of the E-mail message.

- **HTML Editor**

A software program that lets a Web Designer view and edit the hidden html code within a web page. Some modern html editors will both create the code for the designer and modify it as and when required. Very handy if you are a novice Web designer.

- **Hyperlink (or Hypertext Link)**

Those funny little hands that appear like magic whenever you move your mouse over a heading or subject title. Hyperlinks are highlighted text or images which when selected (by clicking the mouse) follow a link to another page or another item within the same page. Hyperlinks can also be used to automatically download such things as sound or video clips, etc.

- **Install**

When you copy an application or program onto the hard drive of your computer you are installing it. This may be done from a CD-Rom, a Floppy Disk, from the Internet (as a downloaded program etc) or from an external device such as a Digital Camera. If your computer program fails at any time you may have to re-install it from either of the sources mentioned.

- **Enterprise Resource Planning (ERP**

The daunting task of planning and organizing business functions in the enterprise. It also refers to a class of software designed to help organizations do this. Often the installation of this type of software can be a complex task, taking many months and many millions of dollars. ERP software can help companies manage everything from sales and marketing to human resources.

- **chkdsk (check disk)**

A Microsoft program that checks your hard drive for logical errors as opposed to physical defects. This program is supported in DOS and all versions of Windows. Windows NT/2000/XP uses it as its main disk checking program. DOS and Windows replaced this program with the friendlier scandisk.exe, but chkdsk.exe is still available and is still the main disk checking program for NTFS partitions.

- **ColdFusion**

A server side extension developed by Allaire that allows documents similar to HTML, usually with the .cfm extension, to be parsed and run on a webserver. ColdFusion allows Web designers to embed database calls into HTML-like documents, and is designed to provide easier access to database servers than by using standard CGI calls.

- **IC (Integrated Circuit)**

A combination of multiple circuits into a single integrated device. Today the common microprocessor uses many millions of transistors, with each transistor counting as a single circuit. This combination is an integrated circuit.

- **IIS (Internet Information Server)**

The name for Microsoft's webserver. It works with server versions of Microsoft's operating systems, and was first developed for Windows NT Server. Starting with Windows 2000 Server, IIS ships on the CD. With Windows NT 4 Server you had to install additional software to get IIS installed.

- **Industry Standard Adapter (ISA)**

The original 8- and 16-bit expansion card standard used in PCs. ISA cards run at a bus speed of 8MHz. You can plug ISA expansion cards into an ISA slot. Modems and sound cards were the last ISA cards due to their low bandwidth requirements. ISA is rarely found on new motherboards nowadays, as it has been replaced by PCI.

- **Information Services (IS)**

This refers to the field of computer technology, but has been replaced by the newer and sexier term "IT."

- **Information Technology (IT)**

The field of work dealing with computers and technology, or more specifically, the organization within a company that takes care of all of the computers, telephones, webservers, and Internet connectivity that keeps a company able to communicate with the outside world by electronic means.

- **Instruction**

The simplest direction that you can give to a processor. Programs are made up of these, but usually don't go down to such nitty-gritty levels unless you are programming in assembly language. An example of what a single instruction would do is increment a piece of data by one, or clear a piece of data out of a register.

- **Integrated Device Electronics (IDE)**

IDE is the standard hard drive interface for PCs. You can connect a maximum of two hard drives to an IDE connection or channel. IDE hard drives are cheaper than SCSI drives, but IDE is generally slower than SCSI and does not support sector re-mapping.

- **Java Servlets (Servlets)**

A Java application that is designed to execute on a webserver instead of on the client's computer. Servlets can be used in the same way that CGI programs can to move data between a client and the webserver.

- **Java Script**

A simple scripting language designed by Netscape to be embedded into HTML documents. It is unrelated to Java. JavaScript is supported in most modern browsers and can be disabled, but it is so ubiquitous at this point that disabling JavaScript will often limit your browsing ability. JavaScript runs on the client, not the server, and is useful for off-loading operations to client machines. However, there is also server-side JavaScript.

- **MODEM (Modulator/Demodulator)**

A device that serves as a bridge between your digital computer and some form of analog line used to transmit data, such as a phone line (standard modem) or analog cable connection (cable modem). The modem can receive the analog signals from the line and turn them digital, or transmit your digital signals into analog signals that are capable of being decoded digitally.

- **Multimedia**

Any use of audio or video in a computer. In simplest terms this refers to the basic functions of sound cards and video cards. The term also covers television and video integration in computers.

- **Multitasking**

The ability of an operating system to run two or more tasks at once. With one processor you will not normally have more than one task using the processor at a given moment in time, but the tasks will be scheduled so that they can all appear to be running at the same time and do not interfere with one another. A task can be a program (e.g., the Windows Calculator) or an instance of a program (e.g., opening the Windows Calculator multiple times).

- **NVRAM (Non-Volatile RAM)**

A typically small amount of RAM that stores information even after you turn off your computer. It is used in modems for storing your settings and in hardware keys for protecting software. Flash memory is a type of non-volatile RAM.

- **NTFS (NT File System)**

This alternative file allocation system available first with the Windows NT operating system, and then with Windows 2000. It allows for larger disk drives with smaller cluster sizes, as well as a performance increase and more robust security. NTFS is now the standard file system for Windows XP.

- **Offline**

When your computer performs an operation and it's not connected to the Internet or any other computer at the time of performing this operation, then it is deemed to be working 'offline'. We can show you how to save the Web pages you want into your Favorites Folder and view them 'offline' any time you want!

- **OR**

Entering the word OR between two words in the Address Bar of your Web Browser will return Web pages that contain either of the two words entered. When typed into a Web Browser between

two other words it's known as a Boolean Expression. Other examples are: AND, NOT and NEAR. Let's say you entered Gold OR Silver - then pages that contain the word Gold OR the word Silver, will be found for you.

- **ODBC (Open DataBase Connectivity)**

A standard API for communicating with database servers. There are different ODBC drivers supporting most of the major database servers, such as Oracle and Microsoft SQL Server. If you program to ODBC you get the advantage of (theoretically) being able to easily use your application on different databases without reprogramming. However, ODBC drivers are not always perfect.

- **OLAP (OnLine Analytical Processing)**

A group of technologies and applications that collect, manage, process, and present multidimensional data for analysis and management purposes. To be effective, results of the analysis done must be presented quickly, and within the same application or a closely linked application.

- **PC (Personal Computer)**

This is slang for IBM Personal Computer, or IBM-PC. This is the class of computers the works (so far) on the x86 instruction set, and were first developed by IBM as a means to put a computer in your home. Before that IBM computers were only used in business. After the PC was developed, many clone PC makers began developing them as well, and that has led to the large amount of components that are PC-compatible; but it has also caused some problems when cheap components try to work properly with one another.

- **PERL (Practical Extraction and Report Language)**

Created by system admin Larry Wall in the mid 1980s, this programming language was originally intended to fill a gap and help out with administration tasks. From those meager beginnings PERL has become a programming language often associated with the Web and UNIX. PERL is the most common programming language used to program CGI scripts. It is a very powerful language that can be used to easily modify large amounts of text files with hardly any programming required. PERL aficionados use it for doing everything.

- **POP3 (Post Office Protocol 3)**

A standard for client/server transmission of e-mail. An e-mail server holds the e-mail, and you use a POP3 client to fetch the mail from a server. IMAP is a newer e-mail client/server protocol with more options.

- **Portable Computer**

Technically, any computer that features a self-contained screen to allow it to easily be moved around. The first "luggable" portables were as big as a suitcase and had small CRT monitors, then came the with orange gas-plasma screens, and then laptops with LCD screens. Portable computers also usually have their own power source. options.

❖ **Portable Operating System Interface (POSIX)**

A set of operating system interface standards based on UNIX. The standards were developed so that programs could be written more easily that would work on multiple versions of UNIX from different manufacturers.

❖ **Portal (Web portal)**

A term coined to describe the large search engine sites, such as Yahoo! and Lycos, that have branched off to offer a wide variety of services. The idea is that a Web user would peer at the Web by using only one website: the portal. For example, you go to a portal to do searches, get stock quotes, buy things, etc. It would be your everything site. Each portal site wants to offer one of each type of service so that a user never has to leave the site.

❖ **SCSI (Small Computer Systems Interface)**

Pronounced "skuzzy," this is a standard data pathway used mostly for hard drives and CD-ROM drives; but it was also a common interface for scanners and even printers at one point. It is the fastest (and the most flexible) method of interfacing with hard drives. It comes in numerous varieties, and is mainly used in servers and high-end workstations. SCSI drives are much more expensive than IDE drives, but SCSI drives can have features, like 15,000 RPM spin speeds and 5-year warranties, that IDE drives currently do not.

❖ **Script**

A group of commands usually stored in a file and run one at a time so that you don't have to type them in one at a time. Script is the newer, sexier term for batch. Don't talk about batch files anymore! It's all scripts and scripting languages. We're on the INTERNET, for goodness' sake!

❖ **Server**

A machine whose sole purpose is to supply data so that other machines can use that data. This also describes any software process that runs on a server machine and responds to client processes or programs locally or across a network. See also client/server.

❖ **Shell**

This most commonly refers to the various text-based user interface programs available for UNIX or Linux. The shell is the part of the OS that interacts with the user and accepts typed commands. Different shells have different functionality, so it is important to have the proper shell loaded or you may find yourself lost as things are displayed differently and familiar commands are not supported.

❖ **Sleep mode**

The placement of a computing device into an inoperable mode, where less power is consumed by shutting down unnecessary devices, but leaving all data in RAM. Typically you return from sleep mode by using the keyboard or mouse, and devices are switched back on. Sleep mode in its early

incarnations was very problematic in some PCs, and would often crash programs and operating systems that were not completely compatible with the sleep mode in the PC's BIOS.

- **SMS (Short Message Service)**

A method of sending text messages that are 160 characters in length or shorter over a mobile phone. More and more mobile phones are supporting the sending and receiving of SMS messages.

- **Software**

This is any computer program, multimedia title or utility that can be purchased from a computer shop and installed on your machine to add extra capability to it. Examples could be an image editing program, a digital camera application, a fast moving computer game or an OCR package. Always check with your supplier that the software you are about to purchase will be compatible with your operating system.

- **Software License**

Most corporations need multiple copies of software, but do not need the media in which they come, either because they already have it or because they allow users to install software from a server on the network. Companies still need to purchase a copy for each user, however, so they need a way to prove they have actually purchased a copy of each. These companies purchase software licenses with no associated media. Such licenses are typically just sheets of paper that cost a lot of money, but allow you to legally use additional copies of the software.

- **Solaris**

A UNIX-based operating system developed by Sun Microsystems and used widely for enterprise-class servers. It is designed to work with Sun's own SPARC chips as well as Intel's x86 microprocessors.

- **Sound Card**

A peripheral device in the form of a card used for producing sound and music. Although PCs come with a built-in speaker, it was originally tied to the CPU and did not do a good job of reproducing complex sounds. As for reproducing sound, the speaker takes most of the processor's power to produce only marginal sound, whereas a sound card does it effortlessly and creates a much better reproduction. Most motherboards today come with some kind of external sound chip built-in, just in case you want decent sound without attaching a separate sound card.

- **SQL (Structured Query Language)**

This is a means of managing data in a relational database. There is a SQL standard, and there are also many vendor-specific SQL packages which combine relational databases with tools SQL tools to manage them. Statements in SQL can be used to read or request data from a database, such as, "select * from geek," which would return an entire table of data from the table named "geek." Queries can also be much more complex such as, "select * from geek where name=sam" which would return records from the database where the field "name" was set to "sam." SQL statements can also be used to delete and update data.

❖ **Spool (n. spool)**

The intermediary device between a computer and a printer. In the old days, if you had no spooler your computer would wait as the printer slowly printed a document. You would send your print data to a spooler to accept the data and save it temporarily to hard disk or memory while it dealt with the slow printer for you. Nowadays print-server is a more current term for describing this type of device. Most modern operating systems contain spooler processes that take care of printing in the background, and you don't notice any delays anyway. Spool

❖ **Stack**

A data construct that uses first-in, last-out (FILO). Think of a stack of pancakes. The first pancake cooked (first in) is put on a plate and then covered with other pancakes as they are done cooking. The original pancake is the last one that leaves the plate if you eat them one at a time. See also queue.

❖ **Standalone**

A hardware device or piece of software that works with nothing else required. Examples include a hardware-based MP3 player, a RAID server that hooks up directly to the network with no PC required to run it, or an executable program with the proper libraries embedded. Standalone can have many contexts, but it always refers to the ability to function without requiring other components.

❖ **Tape Drive**

A device that can store data on a tape. The advantage of storing data on a tape is that a tape can hold large amounts of data in a small and inexpensive package. On the downside, a tape cannot store the data indefinitely, and it is expensive and slower compared to a hard drive. But tapes themselves are cheaper, and are more easy to move around than hard drives.

❖ **Bridge**

A relatively simple device that passes data from one local-area network (LAN) segment to another without changing it. The separate LAN segments that are bridged use the same protocol.

❖ **C**

The programming language created by Dennis Ritchie of Bell Laboratories in 1972 when he and Ken Thompson worked on the UNIX operating system design. It was based on Thompson's B language. It has found widespread use on personal computers and is one of the languages in which expert system shells are implemented.

❖ **C++**

An extension to the C language . As a superset of C, it provides additional features for data abstraction and object-oriented programming.

❖ **A: Drive**

This is another name for the Floppy Disk Drive installed inside your computer. Two letters are normally assigned for use by these type of drives. The first, or primary Floppy Disk Drive is always allocated to A: Drive. If another one was ever fitted in addition to this, it would become the secondary drive and therefore be assigned to B: Drive. C: Drive is of course your primary Hard Drive and is often referred to as your Hard Disk.

- **Accelerator Card**

It does exactly what it says on the packet. A card fitted inside the PC that makes it perform faster, such as a better graphics or sound card with faster memory on it.

- **Access**

The quality of a system incorporating hardware or software that makes it usable by people with one or more physical disabilities, such as restricted mobility, blindness, or deafness.

- **Access Time**

The agreed time when an estranged father gets to see his children. Alternatively, the time it takes for a device to access (i.e. locate) data. It's usually quoted in milliseconds (ms) if referring to a hard disk or nanoseconds (ns) if referring to memory. The computer industry often uses the Access Time as a benchmark (i.e. reference point) when comparing the performance of hard disks or other add-on devices. The lower the (ms) or (ns) the better the performance of the piece of equipment.

- **Active Desktop**

This is a special version of the Windows desktop that will allow such things as Web Pages and 'Live' information to be displayed instead of the usual boring background colours and patterns. To find out if your PC has this feature installed, right click on a clear area of your existing desktop and if Active Desktop is listed in the menu that appears select 'View as Web Page'. Once this is done all you have to do is browse for a suitable Web Page to display and set it up as your wallpaper.

- **Active Partition**

The primary partition that has been set up either by the computer manufacturer or by the main user to be read and used when it boots up (i.e. starts up). It usually contains all the main system files on the computer that you are using now it's most likely on C: Drive - unless you're really posh and have more than one hard drive.

- **Add-On**

An extra bit of hardware that's added to the computer after its purchase in order to improve its performance or capability. Examples are such things as extra ram memory, a better sound or video card, a digital camera, zip drive or web-cam.

- **AND**

This term links two or more search enquiry items together which allows you to narrow down a search. In this context it is known as a Boolean Expression. For example, if you entered the word

Computer AND Tuition you would be presented with Web pages containing both words, and not just either one of them. See also: OR, NOT and NEAR.

- **Animated gif**

A series of individual gif images can be saved within a special animation application so that when they're combined together they form a short sequence of 'what appears to be' moving images. The advertising banners seen at the top of many commercial web pages are often animated gifs, which are designed to catch your attention.

- **ALU (Arithmetic Logic Unit)**

The part of the CPU that actually does the work of adding, subtracting, multiplying, and dividing, including OR, AND, and NOT operations. The ALU is an execution unit, like the FPU, that is fed with data from the CPU registers.

- **Analog**

Analog refers to a representation of a quantity that varies over any continuous range of values. Analog signals can be thought of as pure in nature and not processed. Thus, the debate over whether record albums (analog representation of sound, where the sound is generated by physical ridges on the record) sound better thanCDs (digital representation of sound, where variation is limited by the encoding level). Think of nature as analog. Analog values are exact, but it is impossible to correct errors in reproduction (e.g., a scratch on a record).

- **Array**

1. This can be an abbreviation for a group of hard drives functioning as a RAID (Redundant Array of Inexpensive Drives). 2. This also describes a common data structure that holds values in a consecutively numbered group, such as A[1]...[7], where A is an array of 7 different values. There are also multi-dimensional arrays such as A[1,1,1]...A[8,8,8], which can hold many values and still be fairly simply referenced.

- **Assembly Language**

A programming language specific to a microprocessor. It is a very low-level language, where you actually give the processor instructions like "MOV A,B", which moves a value from one register to another. As you might imagine, programming directly in assembly language is quite tedious. Thus, higher level languages, such as C++, Visual Basic, or Java, are normally used and then compiled into assembly language specific to the microprocessor on which the program will be run. The compiler tries to optimize the code during this process (e.g., "MOV A,B" followed by "MOV B,C" might be replaced by "MOV A,C"). Depending on how elegant the optimization is, the code may run faster than if no optimization is used. Today, very small and fast programs can be created by using assembly language (defeating code bloat), but assembly language programming is becoming a dying art.

- **Embedded Memory**

This is memory that is built directly onto a processor. For example, a graphics chip may have embedded memory instead of using separate memory chips. Use of embedded memory in PCs and PC components nowadays is fairly rare, as attaching a large amount of memory to a chip reduces yields and increases costs.

❖ **Embedded System**

A system that is located entirely on a processor. All logic is contained in a single chip and has a single purpose. New cars have many embedded systems working to keep emissions low and performance high.

❖ **End User License Agreement (EULA)**

The contract found in most software packages that describes the rights to which the user of the software is entitled. Typically, it will explain how many people can use the software, whether it can be used on multiple machines, and whether it is transferable.

❖ **Enhanced SDRAM (ESDRAM)**

A type of SDRAM that includes a small amount of SRAM cache memory for lower latency. It is compatible with SDRAM, but you will not get better performance unless ESDRAM is mated with a memory controller that has ESDRAM support. ESDRAM was never more than a niche product.

❖ **Enterprise**

The entirety of an organization that uses computers. Typically it refers to very large corporations, or software or hardware solutions designed for large organizations. Enterprise companies usually have numerous locations and hundreds, if not thousands, of users to support--a scenario which requires an entirely different type of IT system and management than a small to mid-sized company.

❖ **COM (Component Object Model)**

A Microsoft specification that describes methods of communication between components. For example, if you were to drag an item over a window, the item and window would have some rules of communication that they would follow. Those rules are described in the Component Object Model.

❖ **Common Object Request Broker Architecture (CORBA)**

A standard that allows programs or objects to communicate even though they may have been written by different vendors. CORBA is defined by a group of 800 companies called the Object Management Group, a non-profit consortium that produces and maintains computer industry specifications for interoperable enterprise applications. Microsoft has been a member of OMG since 1992 and has released competing technologies, first with OLE and more recently with DCOM.

❖ **Integrated Development Environment (IDE)**

A development environment that is integrated into an application. For example, many office suite programs feature a macro language that can be expanded by using a common programming language. Specifically, Microsoft Office allows development in Visual Basic for Applications inside of its Office products. Thus, the development environment is integrated into the applications.

- **Java Server Page (JSP)**

A specification that uses Java Servlets run on the webserver to generate Web pages that feature dynamic content. JSP is freely available, and a competitor to Microsoft's ASP.

- **Kernel**

The guts of any operating system. The kernel is loaded into main memory and stays there, while other pieces of the OS are loaded in and out of memory. The kernel controls all requests for disk, processor, or other resources. Generally the smaller and faster the kernel, the faster the operating system will operate. However, larger kernels can provide more functionality.

- **Keyboard**

The main input device on most PCs. It consists of a "board" with a set of buttons on it that represent all the letters in the alphabet, the numbers 0 through 9, and any extra keys, like cursor keys and function keys, that enable some keys to represent additional characters.

- **Keyword**

A term most often used to describe content on a Web page so that search engines can properly index the page. Keywords are not used any longer by most search engines, as they have been abused too many times by people listing keywords that have nothing to do with their pages in an attempt to get extra traffic.

- **Log on (or Log in)**

It's the term that's used to describe what you're doing when you connect your computer (usually via a modem) to a computer Network. You Log On to the Internet (via an ISP) and when you've finished, you Log Off. Sometimes you may have to enter a Password to Log on, or a User ID.

- **L2 cache (Level 2 Cache)**

A piece of fast memory that sits between the L1 cache of the processor and main memory. It is usually larger than L1 cache, and the L1 cache checks the L2 cache before going to main memory for data (unless the L1 and L2 caches are unified--see unified cache). Nowadays L2 caches are almost always on the same die as the microprocessor, but they can be off-chip.

- **License (Software license)**

Most corporations need multiple copies of software, but do not need the media in which they come, either because they already have it or because they allow users to install software from a server on the network. Companies still need to purchase a copy for each user, however, so they need a way to prove they have actually purchased a copy of each. These companies purchase software licenses

with no associated media. Such licenses are typically just sheets of paper that cost a lot of money, but allow you to legally use additional copies of the software.

- **Microcontroller**

This is like a scaled-down computer designed for a very specific task, unlike a desktop computer, which has many uses. An example of an application for a microcontroller would be a traffic light, or the chip that controls the suspension system of your new car.

- **Relative URL**

An URL that does not include strict directory information; instead, you give the link directions like "back two directories and up one directory" in standard command line format like "../../imagedir/image.jpg". This way you can change domain names or IP addresses of your webserver and not have to recode your HTML pages. However, if you change directory names (or depth), you must change even your relative URLs.

- **Runtime Error**

An error that happens when a program is executed. When you run/execute a program and get a runtime error, that means that there is as error in the program that was not or could not be detected by the compiler when the program was initially compiled.

- **Runtime**

The time when a program or process is being executed. When it's running it may need runtime libraries and have runtime variables with runtime values. The term is also used to refer to runtime versions of software that include functionality of the software as the means to an end of running some other software, such as packaging a DOS program with a runtime version of DOS so that you don't even need DOS on your computer to run the program.

- **S.M.A.R.T. (Self-Monitoring Analysis and Reporting**

This technology reports on a variety of hard drive attributes. You need a compliant BIOS and SCSI and/or IDE controller, a hard drive that supports SMART, and some sort of software package that reports on these conditions. Once you have that you should be able to receive system warnings about your hard drive. Many hard drive manufacturers have added onto the SMART technology or changed it around so that it has proprietary features for their drives. The good news about SMART is that having SMART is much better than not having it, and you can be warned of hard drive failure before it happens and backup your drive while it still works. Thus, your data is safer with SMART around.

- **Safe Mode**

An operating mode used in Microsoft operating systems. It was first introduced in Windows 95 and was loaded automatically if Windows 95 crashed during boot up. You can access Safe Mode if you press the "F8" key when new Windows operating systems are booting--this will bring you to a menu that allows you to boot into safe mode. Safe Mode boots the operating system with minimal driver support. The purpose of it is to help resolve boot problems. For example, if you

install the wrong graphics driver, Windows could crash when it's loading. When you restart Windows it will boot into Safe Mode and use the standard VGA driver with 60Hz refresh rate. This will allow you to go to the Display Properties function and switch the video driver back to something that works. There is no reason to go into Safe Mode unless you are crashing during boot-up or you are trying to diagnose a driver problem.

- **Scandisk**

A Microsoft program that first shipped with DOS version 6, replacing the venerable chkdsk.exe program. Technically the program is scandisk.exe. It is available in MS-DOS version 6.x, and non-NT versions of Windows. It added the ability to do a surface scan for physical defects on drive media, and a nicer UI than chkdsk, which had no graphical UI. Windows NT/2000/XP still uses chkdsk.

- **Taskbar**

The space that normally sits at the bottom of the Windows 95/98/NT4/2000/XP interface. It displays the list of running programs so that you can easily switch among programs even when you have a maximized window taking up the entire screen otherwise. It can be moved to either side or the top of the screen.

- **Text Editor**

A class of computer programs that allows the opening, changing, and saving of text files. Text editors can be used to edit HTML files, and any file that is not binary in nature. Text editors are not good for working with graphics files or proprietary formats such as Word documents that contain formatting information that is not translated properly to plain text. Text editors differ from word processors because no formatting data (such as font type, font size, etc.) is added when a file is saved in a text editor.

- **Thread**

Part of a program that runs independently or along with other threads to accomplish a task. To run multiple threads you must be running on an operating system such as UNIX or Windows NT/2000/XP that supports multiple threads. The performance benefit of allowing multiple threads to run at the same time is realized mainly on multi-processing systems. Different threads run on different processors, so they can run simultaneously.

- **Toolbar**

A common user interface term that refers to any rectangular bar of buttons or icons with a set of related functions. For example, most browsers use a toolbar for navigating forward or backwards through pages. You can often customize toolbars and add more functionality to them.

- **Trojan Horse (Trojan)**

A computer program that appears to be something useful, but then does something malicious to your computer. This could range from destroying data to laying dormant and someday hijacking your computer to be used as part of a Denial of Service attack. Anti-virus programs will protect

you from known Trojan horses, but strictly speaking Trojan horses are unlike viruses because they do not replicate. However, combination virus/Trojans can replicate.

❖ **True Colour (24-bit colour) Images**

Images displayed in 24-bit colour (which is also referred to as Photo Realistic Colour) are composed of three 8-bit colour channels. Each one is similar in characteristics to an to an 8-bit greyscale image in that it contains up to 256 colours. When combined, the red, green and blue channels can provide up to a 16.7 million colours. Let us know when you find a Printer that can cope with this many colours - because we want one!

❖ **Thumbnail**

If it is necessary for a Web site is to contain lots of large images, then the best way to present them is as thumbnails. If the user wants to see a particular image they can click on it to make it appear. Most clipart images on CD's are as clickable thumbnails. The dictionary definition is ~ a reduced preview version of an image often used in photographic programs to provide the viewer with a quick browsing experience.

❖ **UML (Unified Modeling Language)**

Initially created at Rational Software (now part of IBM), this is an industry-standard method of specifying, visualizing, constructing, and documenting the artifacts of object-oriented software systems using a graphical diagram that looks similar to a flowchart. You can use UML to effectively make a blueprint of the software you are developing, thus making additional development easier, as you can refer back to your UML model.

❖ **USB (Universal Serial Bus)**

A serial connection technology that is almost universally available in current PCs. Version 1.x allowed for 12Mbps transfer rates, and this was boosted to 480Mbps for USB 2.0. USB 2.0 competes with FireWire for transmission speed. Even though USB is so ubiquitous, the PS/2 port is still used for keyboard and mouse connection on many new PCs.

❖ **VB (Visual Basic)**

A software product developed by Microsoft. Its purpose is to bring programming down to a drag-and-drop level to speed up development cycles. In many ways that goal has been achieved. VB's main competitor at one time was Borland's Delphi. Both programs offer similar functionality, with VB based on the BASIC programming language and Delphi based on Pascal. The actual code generated by VB is BASIC, and you can go in and edit the nitty-gritty if you want to. VB was at one time very slow compared to C++, but it has been sped up significantly since those days.

❖ **VCD (Video CD)**

This technology was developed by Sony and Philips in 1993, and allows around 70 minutes of compressed MPEG-1 video/audio to be stored on a CD. Typically VCD movies are shipped on two CDs. VCDs were very popular in Asia, and were available before DVD. Even though the VCD format was extended with SVCD, VCDs will probably eventually succumb to the higher quality

of DVD. VCD resolution is 352x240 (NTSC) or 352x288 (PAL), which is fairly comparable to VHS resolution of 300x360.

❖ **Virtual LAN (VLAN)**

This is created when a bunch of physically connected ports are grouped together by network hardware that supports VLANs. These VLANs are each treated as completely separate entities, and can only be joined together by a router. This scheme is useful for grouping departments together for security and minimizing network traffic.

❖ **Virtual Machine**

Generally speaking, this is any non-physical construct that runs within the confines of another real (physical) or virtual machine. For example, an operating system is a type of virtual machine that runs on a computer's BIOS software, which runs on a physical computer. Any piece of software that runs on an operating system (or within the confines of another piece of software) can also be considered a virtual machine. The term and idea of a virtual machine has been used by Sun Microsystems in its description of a program launched by a browser that will run Java programs. Virtual machine also describes specific programs that mimic a computer within a computer, or a simulation of a physical device represented by computer software.

❖ **Virtual Reality (VR)**

A world that only exists in a computer, often experienced by looking through 3D goggles that detect which way you are looking and then display what should be there. Another form of virtual reality is a world created in your imagination by stories on the computer, such as a MUD. Someday the computer may be able to plug directly into your brain, giving even more life-like simulations of virtual worlds.

❖ **Virus**

A program that makes copies of itself on the same computer without the user's knowledge. Sometimes these copies are added onto executable files or system files, and other times they are part of Word or Excel documents, called macro viruses. The virus will usually have some eventual effect on systems that are infected, known as the payload. Often the intent of a virus is malicious. Sometimes the intent is not specifically malicious, but due to the spreading of the virus and its use of resources it becomes a malicious act as it causes problems for users.

❖ **Visual C++**

A Microsoft product that is basically VB on steroids. It features a similar visual interface with drag-and-drop functionality, but the code is C++, which is much more robust than BASIC. It's also faster when compiled.

❖ **VPN (Virtual Private Network)**

A "virtual" network constructed by connecting computers together over the Internet and encrypting their communications so that other people cannot understand the communications. The benefit is that people can connect to a local LAN from anywhere on the Internet. This allows easier

connectivity and lower phone bills for travelling salespeople. They just sign up with a national ISP and call local POPs from their hotels as they travel the country, easily connecting back to their company's local network.

❖ **Architecture**

Structure of a part or the entire computer system. Combination of hardware and software linking systems across a network

❖ **Asynchronous**

Characterised by not having a constant time interval between successive bits, characters or events. Transmission generally uses one start and one stop bit for character element synchronisation (often called start-stop transmission).

❖ **Autoexec.bat**

In a DOS system, set of commands executed at each system initialisation (either after power-on or after pressing Control+Alt+Del). This file is restricted to the use of the system administrator. In a PMF system, the AUTOUSER.BAT, executed immediately after the AUTOEXEC.BAT, can be accessed and modified by the user.

❖ **Baud**

Origin of the name : a mister BAUDOT invented many decades ago the telex system, and the associated data coding technique. It measures, for a signal, the speed of state changing. It can correspond to more than the number of bits transmitted per unit of time.

❖ **Baud Rate**

A measure of the speed at which computers send data from one device to another, typically 300, 1,200 or 2,400, with the higher numbers representing faster transmissions. One Baud may carry one or some bits of data per second

❖ **Boot**

On a PC, operation consisting of initialising the system. This is done automatically at power-on, or on request, when pressing the Ctrl+Alt+Del keys. The previous state of the main storage is lost.

❖ **Boot Sector**

Instructions allowing the hard disk to load the DOS operating system. This record is one of the privileged targets for viruses, because it is executed at each computer power-on.

❖ **Bps**

Bits Per Second. Basic unit of measure for serial date transmission capacity ; the number of binary digits transmitted over a communications channel in a second. On a classical dial-up line, this speed was 1,200 in the 70s, 4,800 in the 80s. It is now (1995) commonly 14,400 and 28,800 is arriving. With compression techniques (now common in modems and routers), this speed is

multiplied by a factor 2 to 3, thus allowing a throughput approaching 10,000 characters per second (or more than 30 million characters per hour).

❖ **Binary Format**

Any file format in which information is encoded in some format other than a standard character encoding scheme . A file written in binary format contains information which is not displayable as characters. Software capable of understanding the particular binary format method of encoding information must be used to interpret the information in a binary formatted file. Binary formats are often used to store more information in less space than possible in a character format file. They can also be searched and analyzed more quickly by appropriate software. A file written in binary format could store the number "7" as a binary number (instead of as a character) in as little as 3 bits (i.e., 111), but would more typically use 4 bits (i.e., 0111). Binary formats are not normally portable, however. Software program files are written in binary format. Examples of numeric data files distributed in binary format include: the IBM-binary versions of the Center for Research in Security Prices files, the U.S. Department of Commerce's National Trade Data Bank on CD-ROM . The International Monetary Fund distributes International Financial Statistics in a mixed character-format and binary (packed-decimal)) format. SAS and SPSS store their system files in binary format.

❖ **Attachment**

These are files attached to E-Mail messages, and could be in the form of text, graphics, sound, video, spreadsheet, database or even an entire downloadable application. Larger ones are usually ‘zipped’ first to speed up the time it takes to both send or receive them.

Buffer

A holding area that memorises and stores information, commands, keystrokes, printing requests or CD copying info/data etc, until the computer's processor (CPU) is ready to complete the requested tasks.

❖ **CD-RW**

This is the compact disk re-writable format used on re-usable blank CD's that can be used over and over again.

❖ **CD Writer**

A special unit, fitted into a PC that allows the user to make copies of their own licensed software or music.

❖ **CD Re-Writer**

Similar to the one listed above but capable of erasing the contents of special re-writable CD-Roms and then re-recording onto them.

❖ **Chips**

Not French Fries - but little blocks of silicon with embedded transistor material that process lots of computer information very fast and reliably. Heard of Silicon Valley? Well, that's where 80% of the World's silicon 'chips' are made.

- **Command**

An instruction given to the computer by the user which is expected to be carried out.

- **Compression**

The pressure above a piston in a petrol engine when it's coming up to the point of ignition. Also the squeezing of computer files in such programs as DriveSpace in order to free up much needed hard disk space.

- **Control Panel**

Most of your Windows Operating System Settings can be viewed from here as Icons. By double-clicking any of the icons displayed here you will be able to access most of your computer's functions and provided you know what you're doing, make several changes to them.

- **Cookies**

American name for a sweet cake, bun or biscuit. Also known as a data file that a Web Server sends to your browser when you visit a Web site. The cookie is updated each time you return and holds info about you, which may be used later. Let's say you are browsing around an Internet Virtual Shop, each time you place something in your basket the info will be stored by the cookie. When you eventually decide to buy the items in your trolley it's the cookie job to supply the info to the server so that the order can be processed. Beware, some web sites use cookies without asking your permission. See us if you want to know how to set up your browser to prompt you when this is happening.

- **Alpha Test (Alpha)**

A term given to a very early version of a hardware or software product which is not yet stable and may lack features. After some testing and some revision, the product will assume beta status

- **Arrow keys**

The four keys on a computer keyboard that are commonly used to move the cursor around in programs that support such movement. Some keyboards feature 8 arrow keys, which include the four standard directions (up, down, left, right) and the four diagonals.

- **AT Attachment (ATA)**

A storage (hard drive, CD-ROM, etc.) interface more commonly known as IDE. The "AT" refers to the IBM-AT computer where this interface was first used.

- **Back-end**

A type of program or process that is not directly accessed by a user. Often it will carry out its tasks independently of the front-end or user interface. For example, a user could request data from a database, not knowing that the data is refreshed on the back-end on a daily basis.

- **C Sharp (C#)**

An object-oriented programming language from Microsoft based on C/C++ that contains functionality similar to that found in the Java programming language. It is designed to work with Microsoft's .NET platform.

- **Circuit**

Most commonly, this describes an electrical device with a defined path of electrical current that can receive input voltages in a 0 range and a 1 range, and responds with an output voltage that is also in a 0 or 1 range based on the logic inside of the circuit. If the circuit has a 0 range of 0-1 volts and a 1 range of 4-5 volts and receives a 0.5 volt input, it will act as if it has received a "0" input. Ranges are necessary because voltages are never exact. When thinking about the logic behind a circuit, it is easiest to think of the inputs and outputs simply as a 0 or 1 instead of a range of voltages. See also integrated circuit

- **Client/Server**

Client server technology came about when computers began to cost less. Mainframes are very expensive, and didn't give users much personal freedom. The client/server model promised to change that scenario, and it's much more popular today. Basically, a client computer with its own memory and hard drive communicates with a server whenever it needs data from the server. The client can run by itself without the server and communicate with different servers as it needs to.

- **Clock Cycle**

Think of a clock cycle as one tick of the second hand (but generally at a much higher speed). Computer clocks run voltage through a tiny crystal that oscillates at a predictable speed to give a meaningful timing method to the computer. One clock cycle doesn't necessarily mean that the processor does one operation. Today's high-end processors often complete more than one operation per clock cycle, and other times, in the worst cases, it will take several clock cycles to complete one operation.

- **Clock Speed**

The speed in MHz of a microprocessor. It is one way of gauging the performance of a microprocessor; however, different processor architectures dictate that a higher clock speed in one architecture does not always mean better performance over a lower clock speed in another architecture.

- **CMYK**

An alternate color scheme to the RGB color scheme. Combinations of cyan, magenta, yellow, and black are used to represent colors. The CMYK scheme is used mainly in print, such as magazines. Combining cyan, magenta, and yellow produces black, but that black is not always pure enough,

thus, the addition of the K, for pure black. Color inkjet printers use CMYK to represent images. The best printers have separate black instead of wasting all the colors to print a faux brownish black.

- **COBOL (COmmon Business-Oriented Language)**

A programming language developed in the '60s by several computer companies and the U.S. Department of Defense. COBOL is still used today for programming business applications, and COBOL programmers were a major source of the Year 2000 headache. In fact, many of them came out of retirement to fix the mess they made, whether voluntarily or by direction, to save a couple of valuable bits of data back when bits cost big money.

- **Command Prompt**

Any blinking cursor waiting, or prompting, for user input. In DOS the C: prompt greets you on most systems--this is a type of command prompt. As well, if you use any version of Windows you can get to a DOS-looking window that allows you to type in commands. UNIX can also greet you with a command prompt. For novice users a command prompt can be confusing, as it's unclear what to do next; but for experts a command prompt is a necessity at times.

- **Compiler**

A compiler translates a computer program from one language into another, catching any errors in syntax along the way.Most commonly, you translate some high level language, such as C++ or COBOL, into optimized machine language. This form of compilation puts your programs into a form that your computer (specifically your microprocessor) can understand without any translation, thus speeding them up greatly over programs that must be interpreted as they are run.

- **Complex Instruction Set Computer (CISC)**

Microchips that support a large amount of instructions of varying length. On the other side of the coin, you have RISC chips that use a smaller instruction set of more regular length. The battle between RISC and CISC rages on, but when it comes down to it, either way has its advantages and disadvantages. With CISC you can implement often-used strings of instructions into single instructions that could give you an advantage over the numerous small instructions used in RISC. However, the unpredictability of the length of a CISC instruction typically limits CISC from blowing RISC away.

- **Conditional Statement**

In programming, this is a type of command that controls the flow of a program based on whether certain conditions are met. The statement is normally set up in the form: IF (specific condition is met) THEN (do this action) ELSE (do this if the condition is not met). The "Else" statement or condition does not usually have to be listed specifically. It can be implied that if there is no type of Else statement then program flow continues normally if the condition is not met. Conditions can range from simple (x equals 3) to more complex conditions (x is not less than or equal to the number of characters in a string).

❖ **Cracker**

This is the common term used to describe a malicious hacker, though it also can refer to code breakers. Crackers get into all kinds of mischief, including breaking or "cracking" copy protection on software programs, breaking into systems and causing harm, changing data, or stealing. Hackers largely regard crackers as a less educated group of individuals who cannot truly create their own work, and simply steal other people's work to cause mischief or for personal gain, not to promote understanding.

❖ **Ctrl (Control Key)**

A key on a computer keyboard that typically adds 64 bits to the ASCII value of a key being pressed. Based on the program that is running, it can have different effects. It was added to create more key combinations besides just using the Shift key. In selection of items, holding down the Ctrl key will typically allow you to select or de-select a single item from a group without affecting the rest of your selection.

❖ **Cylinder**

This term is somewhat synonymous with the tracks on a hard disk drive. However, instead of a single track, a cylinder refers to the location of all the drive read and write heads, typically accessing multiple platters. Thus, when these tracks are mentally pictured, they are stacked up like a cylinder. Since all the heads are locked together, a specific cylinder number is equivalent to a track number on a specific platter, but refers to all platters at once. Got it? Whew.

Data

Any type of information that is created or changed whilst working on the computer.

❖ **Data Transfer Rate**

The speed at which data bits are transferred along a transmission channel. The data transfer rate is usually measured in bits, kilobits or megabits per second.

❖ **Default**

One of the most misunderstood words in the Internet Dictionary. When a computer program is designed several user options may be available, so to simplify matters the designer will set the one that he thinks will suit most people's requirements - as the Default Option. This can of course be changed if it is not to your liking. For example in MS Word the Default Font is usually 12 pts Times New Roman. You might prefer 10 pts Ariel. OK, let's change it. Go up to Format, select Font, select 10 pts Ariel, click Default, confirm the change and click OK. From now on every time you open MS Word your font setting will be 10 pts Ariel. This is now the new Default Setting.

❖ **Desktop**

This used to be what we called the top of a desk, but not anymore. It's the popular name for the Windows screen that you see when your computer has settled after starting up. Desktop Shortcuts are usually displayed on it, as well as the Taskbar (usually at the bottom) and the System Tray

(usually on the right near the clock (if it is enabled). If Active Desktop is available on your computer, you will be able to select moving images, web pages or animated backgrounds to be displayed.

❖ **Dialogue Box**

This is the box that often appears in Windows to display warnings or messages telling you about what's going on. Usually there's a cancel button and an OK button as standard, plus other options if relevant. You must make a choice to continue.

❖ **Domain Name Server**

This is a special computer connected to the Internet whose job it is to keep track of the IP Addresses and Domain Names of other computers. If required, it can take the ASCII Domain Name and convert it into the relevant numeric IP Address.

❖ **E-commerce**

The selling of either goods or services by advertising on the Internet. The modern way of doing business.

❖ **E-mail (Electronic Mail)**

A method of communicating with other persons by sending and receiving electronic mail messages via an ISP's server to other locations for the attention of another computer user at a specific E-mail Address.

❖ **exe file**

This is a program file, also known as an executable file which usually makes something happen!

❖ **Favourites**

This is a folder that contains a list of web sites. It saves you the bother of having to retype complicated internet addresses every time you want to revisit the web address. You can add, arrange, rename or delete web addresses from your favourites folder any time you wish.

❖ **Field**

Several well known Search Engines allow you to specify a 'Field' when conducting a search enquiry. It could be that you want to specify a date, a certain domain, or a specific area of interest. They will then proceed to look through their records for the information that you require using the 'Fields' that you have selected as a guideline in their search.

❖ **Filename Extension**

This is the suffix (i.e. the letters after the dot) in a file name. Examples include .doc (MS Word) .xls (MS Excel) .txt (Notepad) and .pub (MS Publisher). Without a file extension at the end of the file name, Windows will not know what application to launch in order to display the file.

❖ **File Server (or Server)**

A program running on a network that stores files and provides access to them. Web sites are uploaded as files to a File Server and thereafter the Web Designer has (password) access to them to update them as and when required. Everyone else has access to view them by typing the web address into their web browser when online.

❖ **Floppy Disk Drive**

All modern computers have one and it's usually referred to as 'A' Drive. The specifications for a Floppy Disk used in it are: 3.5in HD (high density) and 1.44 MB (capacity). * Always buy Formatted Floppy Disks and keep them in a clean dry place well away from any magnetic source such as a speaker, or an other electrical device.

❖ **Font**

A collection of characters with pre-defined sizes such as the favorite one used on the Web (i.e. Times New Roman) which is the one you are reading right now. The text in a document can be selected and the font size, type or colour changed according to your own personal preference. Most good computers offer the user at least 100 fonts.

❖ **Forms**

Certain Browsers support electronic mail-forms which can be filled in by the users all over the world and the information sent electronically back to the relevant domain site or server that requires the information.

❖ **Frames**

Often used in a Web page to divide it up into sections. Sometimes you'll see an index displayed on the left-hand side of the page and then images or a drop-down menu on the right-hand side, with text in between the two. To achieve this effect a Frame will have been used. There are lots of different types available to a Web Designer.

❖ **Function**

Now this is an in-built calculation such as Max, Sum, Average and If - often used in a spreadsheet. For example: Max finds the highest figure and Min finds the lowest figure in a Microsoft Excel spreadsheet.

❖ **Fifty Six K [56K]**

Abbreviation for a 56,000 kbps modem. It's the speed at which many modems run. At least that's the theory. Now comes the reality. Right-click the two green TV's (next to your clock) next time you're online and select Status. Take a look at your modem connection speed and I'll bet it's nowhere near 56k!

❖ **GB**

One thousand megabytes

❖ **Hardware**

Any physical part of a computer system that you can rap your knuckles on. Eg: monitor, keyboard, mouse, printer, scanner, camera, speakers or external plug-in device.

- **Icon**

A picture or symbol displayed in the form of a graphic image on pop-down menus, toolbars or on your desktop, etc to identify and activate (if clicked on) an application, computer program, command, file or event, etc.

- **Image Map**

An image with clickable 'hot spots' which allow several hyperlinks from a single image file. An example would be an image of a country, split into different areas, each of which could be clickable and hyperlinked to a larger view of that specific area.

- **EBCDIC (Extended Binary Coded Decimal Interchange**

A way of encoding 256 characters in binary, much like ASCII, but used mainly on mainframes. Most of the time EBCDIC is only mentioned in translations between EBCDIC and ASCII.

- **EBIOS (Enhanced BIOS)**

This translates between the partition table limitations of a standard computer BIOS and the IDE limitations to provide up to 8 GB of storage space using the IDE interface. Your computer's BIOS has maximums of 1024 cylinders, 256 heads, and 63 sectors (8 GB). The IDE interface has a maximum of 65,536 cylinders, 16 heads, and 256 sectors (128 GB). Put these maximums together (1024 cylinders, 16 heads, and 63 sectors) and you've got a measly 504 MB of data to work with. The EBIOS translates these limitations in such a way that you can actually achieve the BIOS max of 8 GB on one IDE device. Newer IDE standards have since been developed to up the top hard drive size to 128 GB and beyond.

- **EIDE (Enhanced IDE)**

This standard allowed for two IDE channels that can each support two devices in peaceful coexistence. It also allowed for hard drives up to 8 GB in size and CD-ROM devices

- **EISA (Extended Industry Standard Architecture)**

This is the 32-bit extension of the 16-bit ISA expansion slot. It was generally used only in server machines, and it never caught on for consumers because 32- bit VL-Bus and PCI were more accessible. However, for server machines, EISA beat out VL-Bus and PCI because you could have numerous EISA slots in one machine without any complex wiring. Some server boxes had as many as 14 EISA slots. VL-Bus and PCI initially supported fewer than four slots, and that was a problem if you wanted to install 7 network adapters and 5 RAID cards. EISA made it possible. Nowadays PCI is much more robust, and VL-Bus has long since fallen by the wayside.

- **Embedded Processor**

A microprocessor used in an embedded system. Typically these processors are smaller, consume less power, and utilize a surface mount form factor, as opposed to more standard consumer

processors. Embedded processors are only sold to consumers pre-built into embedded systems, not separately.

- **Emoticons**

These are groups of text characters that, when viewed sideways, look like facial expressions. For example, the most common one is the happy face :) -- a colon and a close parenthesis. There are many, many more, such as the semicolon wink ;) or the colon-p :P that looks like you are sticking your tongue out. The purpose of emoticons is to convey a feeling in a text-based message that may or may not be obvious to the reader. They are typically used in informal communications, especially when using sarcasm or trying to add extra feeling to the message.

- **Encrypt (v. to encrypt)**

The act of making data unreadable in an orderly fashion so that it can be decrypted later.

- **CAD (Computer Aided Design)**

Oh, you cad! This refers to the use of computers to design things. There are specific CAD programs like AutoCAD that are generally resource-intensive, requiring fast processors, lots of memory, and a big, clear monitor for best results. CAD has enabled people to easily model, create, and walk through or view designs of 3D objects or floor plans from different angles on a computer without actually taking the time to make a physical mock up. This is a huge time saver, and has revolutionized design in general.

- **CDMA (Code Division Multiple Access)**

A 2G digital wireless technology that allows multiple calls to share a radio frequency 1.23MHz wide in the 800MHz-1.9GHz band without causing interference. This is accomplished by assigning each call a unique code and varying its signal by that code to allow only the caller and receiver with that code to communicate with each other. The original CDMA standard allows transmission of up to 14.4Kbps per channel, with up to 8 channels being able to be utilized at once for 115Kbps speeds.

- **CDMA 2000 (Code Division Multiple Access)**

The multiplexed version of the IMT-2000 standard developed by the ITU, and it's part of 3G wireless technology. It increases wireless data transmission speeds of the original CDMA standard to 144Kbps using a single channel and 2Mbps by utilizing 16 channels.

- **CRM (Customer Relationship Management)**

A class of enterprise software that enables a large company to manage all contact (or "touches") that it has with its customers. It would track, for example, calls to tech support, faxes, e-mails, direct mail, telephone contacts, and any other contact that a company would have with a customer and vice-versa. This information can be used for analysis of customer relationships, and gives salespeople an understanding of what to say when calling up a customer.

- **Errata**

Minor errors in microprocessor design that are corrected with a new stepping of the processor. The difference between errata and a serious defect is really based on how many people or systems are seriously affected by the issue. It is standard practice for the first volume production of microprocessors to contain errata that is fixed in later generations. Most early errata are worked around by chipsets, BIOS, and drivers.

- **Error**

This occurs in a program when it encounters a situation that it was not programmed to deal with. If errors are trapped properly, they can be dealt with by the program itself. If not, they will be dealt with by the operating system on which the program is running.

- **Execution Unit**

The part of a microprocessor pipeline that actually follows and runs the instructions that are sent to the CPU after the instructions are decoded.

- **Export**

When you export data you are taking that data from a program, database, or file and saving it in another format that is generally easier to manipulate or pull into a different program. An example would be pulling data from a SQL database and saving it as text so that you can use it in a mailmerge. Thus, the exporting frees the mailmerge program from having to understand the complex SQL format--it just needs to understand the exported text file.

Extensible Markup Language (XML)

A standard created by the W3C. It is a language with many similarities to HTML. What XML adds is the ability to define custom tags, such as , and define the meaning of those tags within the XML document itself--thus the term "extensible." You can extend the XML language easily. XML is becoming more and more common as more browsers and webservers support it. It is also a very flexible way to exchange data over the Web and interpret and use data from other websites.

- **Extension (File Extension)**

DOS, Windows, and, to a lesser extent, UNIX and Linux use the last three characters of a filename, after a period, to signify what type of file a file is, such as text, movie, sound, etc. The MacOS has a file system that contains an identifier for file types that is invisible when looking at the filename, and allows you to call files whatever you want without a messy extension on the end. Some common examples are: .exe (executable file), .bat (batch file), .wav (wave sound file), and .txt (text file).

- **Fat Client**

Today's fast PCs are fat clients. They've got lots of memory and big hard drives. They store information and typically run programs locally off of their hard drives. Fat clients usually work in client/server environments, where they can take on some of the processing workload and leave the servers free to serve data. Fat clients are much more customizable and powerful than thin clients,

but they are harder to manage and control. For example, users can't install AOL software on a thin client.

❖ **FC-PGA (Flip Chip-Plastic Grid Array)**

This is Intel's newer packaging of the Socket 370 design. It features a different electrical setup than Socket 370, but is physically compatible. Thus, old Socket 370 motherboards will not be compatible with new FC-PGA chips, but new FC- PGA motherboards may be able to handle Socket 370 processors. It is not clear why Intel made the electrical change. The physical changes put the core closer to the surface, allowing better cooling as the processor core comes in closer contact to the heatsink.

❖ **FDisk**

The program Microsoft operating systems MS-DOS and non-NT versions of Windows use to create partitions on hard drives. Technically, the program is called fdisk.exe. It uses a text-based interface. Windows 95b first added support for FAT-32 partitions into fdisk. Before that it only supported partitions up to 2 GB using FAT-16. This is also a slang term for wiping a drive out completely, as in "I'm going to F-Disk this drive if Windows crashes one more time!" There are several non-Microsoft equivalents to fdisk, but all serve similar purposes--to allow partitioning of hard disk drives.

❖ **Fiber Optic**

A method of physical data transmission that is a newer alternative to sending electrical signals over copper wires. The way it works is by pulsing light down a strand of glass. These pulses represent binary code--so far that's no better than copper. The advantage is that a single strand of fiber optic can carry thousands and thousands of different frequencies at once without data loss.

❖ **FIFO Buffer (First In First Out Buffer)**

An area of memory that holds information in the order in which it was received until the computer has time to use it.

❖ **Flat File**

A database that contains a single table and can be easily represented using plain text. This type of database contrasts with a relational database, which can contain any number of tables that are linked together. Often, to keep things simple when transferring data between organizations, people will request a flat file. Common flat file text formats include tab-delimited and CSV.

❖ **Flat Screen**

This typically refers to a CRT monitor that is made more flat than a standard tube by using more than one electron gun. It is most useful to professionals who rely on the precision of their monitors. It is also better for the standard user because a straight light across the monitor looks straight and not curved. This can also refer to flat panels and projection screens, which are also flat.

❖ **Floating Point**

A three-part representation of a number that contains a decimal point. The number is represented first by the sign, then the number itself, then decimal position. Some examples of floating point numbers are 4.23423412, 1234.1234234, or 4.00. Floating point numbers offer a specific amount of precision, often 8-bit, 16-bit, 32-bit, or 64-bit. This precision controls how accurately floating point results are represented and calculated during arithmetic operations between floating point numbers. For a simple example, if you have a low level of precision and you divide 1 by 3, you will get 0.33. With a higher level of precision you would see that it is 0.333333333333, and so forth. Thus, your calculation is inaccurate by 0.003333333333 (which is 0.333333333333 - 0.33). A small inaccuracy such as that may not matter if you are pumping out frame rates on a 3D game, but if you are building an airplane's engine you might want to make sure your design program handles a proper amount of precision for the job you are doing. When writing software, larger floating point precision takes more space to store, and may be slower depending on the hardware you are running on.

- **Fortran**

A high-level programming language, a bit more advanced than BASIC but not quite as complex as C. This language refuses to die because it is so huge in the scientific research community. It's not a tough language to learn, and it's fairly powerful. About 60% of scientific programming is still done in Fortran.

- **Frame Relay**

A packet-based communications method for connecting networks. Nowadays it is commonly used to interconnect remote offices over the Internet or private LANs. Frame relay has no error checking, and assumes that devices on either side will be able to check for errors themselves. A frame relay connection can use an ISDN line for slow speeds, or can be over a T1 line or better if faster speeds are needed. To get a frame relay line installed you contact your phone company or Internet Service Provider.

- **Frames Per Second (FPS)**

This term relates to video or 3D games, and is the amount of single full screen (or full window) pictures (or frames) that are displayed each second to generate what appears to be a moving image. Typical digital video displays at around 30 FPS. Higher frame rates are not really necessary, as the human eye cannot typically handle more than about 30 FPS. Some games will average 30 FPS or higher and look choppy. Usually this is due to intense scenes that cause the action to dip below 30 FPS for short periods of time. Rates of over 100 FPS in games are common nowadays with less complex 3D action games, but do not necessarily add to the experience of playing a game.

- **Front Side Bus (FSB**

The speed of the bus connecting the microprocessor, its chipset, and connected main memory. In architectures where the processor interacts directly with main memory, the definition of a singular front side bus is less clear. In such a case you would have to specify two FSB speeds, one for the connection to main memory and one for the connection to the processor chipset.

❖ **Full Duplex**

Originally this referred to a communication between a modem and a remote system, where characters were sent both ways over the phone line so that they could be accurately displayed on a terminal. Now full duplex has taken on the meaning that signals can be sent in both directions at the same time, such as in network communications. This either requires twice the amount of wires or differing frequencies for each type of signal so they do not interfere when on the same wire. Full duplex network connections are preferred, especially for servers, which must send and receive a lot of data.

❖ **Function key (F1, F2, etc.)**

One of the set of 12 keys at the top of a standard computer keyboard. These keys are labelled F1 through F12. The keys are basically general purpose extra keys so that programmers can assign the keys to special functions in their programs. One handy and common use of F3 in applications is to "Find again," or find the value again for which you most recently searched.

❖ **Fuzzy Logic**

Logic without an absolute true or false. Instead, you have gradients of true and false. This is necessary for solving some problems, especially those involving artificial intelligence. For example, the question, "Do I get some food now?" isn't always yes or no, and varies due to environmental factors and degrees of hunger.

❖ **Garbage Collection**

A process where dynamically allocated blocks of memory are reclaimed while a program executes. Garbage collection is normally performed by the garbage collector, which is part of the runtime system. However, it is possible to explicitly write code to do garbage collection as well. Automatic garbage collection is triggered when the amount of free memory blocks falls below a certain threshold.

❖ **Gate**

A gate is a tiny electronic switch. These switches, when linked together, can perform logical functions. Basically, gate is the logical term for transistor.

❖ **Genetic Algorithm**

An algorithm that uses fuzzy logic and can refine itself based on its ability to select proper answers. Often, a human must tell the algorithm what it did right and what it did wrong so that it can select better results the next time it is run.

❖ **Geographic Information System (GIS)**

A system for capturing and manipulating data relating to the Earth. A common use of GIS is to overlay several types of maps (for example, train routes, elevation data, street maps) to determine useful data about a given geographic area.

❖ **Global Positioning System (GPS)**

A system of satellites around the Earth that broadcast the time via radio signals based on an internal atomic clock. GPS devices can receive the signals from multiple satellites, and by measuring the time it took the signal to arrive they can determine your current position on the Earth.

- **Graphics**

The pictures that computers display. Not text, but just about everything else.

- **Hacker**

Someone who seeks to understand computer, phone, or other systems strictly for the satisfaction of having that knowledge. Hackers wonder how things work, and have an incredible curiosity. Hackers will sometimes do questionable legal things, such as breaking into systems, but they generally will not cause harm once they break in. Contrast a hacker to the term cracker or malicious hacker.

- **Hibernate**

This allows computer users to save the contents of their computer's memory to disk before shutting down the PC. When restarted, the contents are read back into RAM and the computer is brought back to the exact state it was in before hibernation was initiated. This is different from sleep mode, where the computer is not fully shut down.

- **HTTPS (Secure HyperText Transfer Protocol)**

A secure means of transferring data using the HTTP protocol. Typically HTTP data is sent over TCP/IP port 80, but HTTPS data is sent over port 443. This standard was developed by Netscape for secure transactions, and uses 40-bit encryption ("weak" encryption) or 128-bit ("strong" encryption). If you are at a secure site, you will notice that there is a closed lock icon on the bottom area of your Navigator or IE browser. The HTTPS standard supports certificates. A webserver operator must get a digital certificate from a third-party certificate provider that ensures that the webserver in question is valid. This certificate gets installed on the webserver, and verifies for a period of a year that that server is a proper secure server.

- **Hypertext Preprocessor (PHP)**

Created in 1994, this is a versatile embedded scripting language that can be placed into HTML documents. As long as the webserver supports it, PHP can be used to generate HTML pages by accessing a database. PHP code is executed on the server, and offers an alternative to CGI or SSI calls, or the use of languages such as ColdFusion.

- **Link**

Links are the connections between hypertext pages. Every time you click on highlighted text to go to another page you are following a link. They often appear as a ‘hand’. The A-Z at the top of this page provides Links to all of the sections within it and Links to the top of the page and the E-mail program.

- **L1 cache (Level 1 Cache)**

A small piece of very fast memory that's almost always on the CPU chip itself. It sits between the CPU registers and the L2 cache. Typically L1 cache has a lower latency than L2 cache, making it more expensive to produce and harder to produce in larger quantities without additional complexity.

- **Link (hyperlink)**

Part of an HTML document that points to another resource. When you view an HTML document using a browser, it is common practice to display hyperlinks in blue with an underlined font. When you click on a hyperlink you will jump, or link, to another area in that document or a different document. The linked document or item may be on the same page, the same server, or a server hundreds of miles away. The work all goes on behind the scenes as long as you are connected to the Internet.

- **Mainframe**

Basically a large and powerful computer designed to be very fault tolerant. Historically, mainframes with lots of memory and disk space are hooked to a bunch of dumb terminals that can be used to access data and run programs on the mainframe, but can do nothing without the mainframe. See also Client/Server.

- **Management Information Systems/Services (MIS)**

The department at most companies that everyone loves to hate. MIS people are the people who work with Information Technology, now more commonly referred to as the IT department.

- **Master**

When two IDE or EIDE devices are put on the same cable, one must be master and the other slave. The master/slave configuration is used not only to allow communications to work properly for two devices on one channel, but also so that there is a specific boot order when two or more IDE hard drives are encountered on a system. The master hard drive on the first IDE channel will be the first IDE drive checked for a master boot record when the system attempts to boot from an IDE device.

- **Master Boot Record (MBR)**

The first sector on a hard disk or other disk media. When a computer boots up it searches for a master boot record wherever the BIOS tells it to (usually the master hard drive on the first IDE channel, but this can also be checked for in CD/DVD drives and floppy drives) and, based on what the master boot record says, loads up an operating system. Thus, if the master boot record becomes corrupt or is tampered with by a virus, it can cause your computer to be unable to boot.

- **Matrix Math Extensions (MMX)**

64 additional instructions for matrix math operations that are commonly used to process multimedia data. This was a slight improvement of the Pentium chip design that was supposed to make it easier and faster to deal with video and audio. The improvements weren't very great, however, and this led Intel to develop the SSE instruction sets.

❖ **MB**

This is roughly one million bytes. This is exactly 1,048,576 bytes (that's 1024 x 1024, or 2^20).also an abbreviation for Motherboard.

❖ **Memory**

The ability to remember things on a computer is known as Memory (or RAM) which stands for Random Access Memory. The difference between human memory and computer memory is that in a computer it's only a temporary storage area which is emptied and lost when the computer is switched off. Some might say that when we get old our human memory become like ram memory! It's measured in Megabytes and usually, 128 Mb is ok- but 256 is so much better.

❖ **MCSE (Microsoft Certified System Engineer)**

A certification program offered by Microsoft. To become an MCSE you must pass a series of standardized tests at Microsoft-approved testing facilities. On the positive side, the MCSE certification is designed to give employers an easier way of screening candidates. On the negative side, a candidate with an MCSE and little practical knowledge may unfairly receive consideration over a more qualified non-MCSE. Some Geeks think it's worthy to aspire to be an MCSE, but others think it's just a waste of time and money to learn the "Microsoft way."

❖ **Megahertz (MHz)**

One million hertz, or one million cycles per second.

❖ **Minicomputer**

This term refers to "mini" mainframe computers that are a step smaller than a large corporate mainframe. They were once popular in small businesses that couldn't afford real mainframes, but now you only find reference to them in free subscription surveys for computer industry print magazines ("Do you work with mini-computers, microcomputers, etc.?") or in some college campuses or large business networks.

❖ **Multiplexer (Mux)**

A logic circuit that sends one of several inputs out over a single output channel. In the network world it is used to describe devices that send several signals over a single line at the same time. The device on the other side of the wire that receives the signal is a demultiplexer.

❖ **Network**

A group of interconnected computers. The computers must be capable of transferring data to form a true network--you can't just weld a bunch of computers together. Put that torch down!

❖ **NOT**

By including this term into the address bar of your browser the search enquiry will only find the word preceding NOT. For example ~ If looking for let's say: watches NOT clocks, you will be

presented with Web pages that contain the word watches but pages that contain both words will not appear.

- **Normalize**

A verb used to describe what can be done to data to remove useless or extraneous entries. For example, if you set up a survey with choices A, B, and No Response, and then wanted to report the % of respondents that picked A or B, you could cut out the "No Responses" and thus normalize the data. Another example would be changing the loudness of an MP3 file by analyzing the file for the loudest part and then setting the loudness of the entire file to a percentage of that. That way, if you are putting together a mix file from many separate sources, you get a more uniform loudness.

- **Radio Button**

A GUI term denoting that the user has a group of selections to make, and that he or she can only make one selection at a time. As it relates loosely to a radio, you can only listen to one station at a time. In the computer world, an example of where a radio button may be used is on a website where you need to pick which type of payment you're using: Visa, Mastercard, Discover, or AMEX. Usually you'll have radio buttons, and you can pick only one method of payment. Radio buttons are represented by a group of small circles. When you click on one of them you get a dot on your selection. There is always a default selection.

- **NOS (Network Operating System)**

An operating system designed to run across a network. It refers to the operating system that runs on a server, not the client. Network OSes are typically designed to provide access to server resources to clients, making the server function as a file server, print server, or other type of server. back into your computer. You will have to reformat your hard disk drive and re-install Windows.

- **Rambus In-line Memory Module (RIMM)**

The form factor for Rambus RDRAM. By comparison, SDRAM is mainly found on DIMMs, and EDO RAM is usually on SIMMs. RIMMs require that if you do not fill all RIMM slots with RDRAM memory you must keep the empty slots filled with termination boards to ensure that the high speed Rambus memory signals do not bounce improperly.

- **Recommended Standard 232 (RS-232)**

This is the de facto standard for communication through PC serial ports. It can refer to cables and ports that support the RS-232 standard. Common usage would include, "Hey, Jimmy! Why don't you take your RS-232 cable and stick it right in your RS-232 port!"

- **Reduced Instruction Set Computing (RISC)**

This type of chip use a simpler instruction set than CISC chips to get its work done. This results in more instructions that need to be processed by the processor, but they are easier to process and regular in size, so the chips can process more instructions per clock cycle than a CISC chip. Chip philosophers argue the benefits of RISC vs. CISC, but there is no clear cut winner. See also CISC for additional info.

❖ **RGB (Red Green Blue)**

CRT monitors, including television sets, use red, green, and blue light to represent the entire color spectrum. When you put red, green, and blue light together you get white light--this is ideal for representing images on the black screen. Most graphics programs will let you create colors by setting levels of red, green, and blue. See also CMYK.

❖ **Robot**

Besides being a mechanical device used to mimic human form, usually to accomplish some repetitive task, this term refers to a computer program that scans Web pages and links. Like a similar spider program, robots are used to scan Web pages and index them. You can insert a file called robots.txt to the main directory of your website to tell the robots which directories not to index.

❖ **Root directory**

The base directory of an operating system. This term is typically used for the UNIX OS, but can apply to a webserver as the directory to which a browser defaults. It refers to the directory represented by a slash, or "/", character in UNIX, or a "\" character in Windows. To change to the root directory use the command "cd /" in UNIX or "cd \" in Windows.

❖ **Save (v. to save)**

This term describes the movement of data from a computer's volatile DRAM to the non-volatile hard disk or other media. Basically, when you save something you are making sure that if your computer loses power you will be able to get back to the information that you have saved. This information may be saved to a floppy disk, a hard drive, or a CD-R. The information also can be saved to these devices connected to your local machine, or to a server on a network.

❖ **Unicode**

A universal encoding scheme for characters and text. The goal of unicode is to enable the use of all characters for all languages of the world. Unicode supports a 16-bit character code for a possible 65,000 characters, as well as an extension of the 16-bit code called UTF-16 that allows for over one million characters and is sufficient for all known encoding requirements, including all known past and present written language characters in the world. Contrast that to the 8-bit 256 character limit of ASCII. Unicode is the default encoding of HTML and XML.

❖ **Uninstall**

When you uninstall a program or application, you remove it and all of it's associated setup or data files from your computer. The safest way to do this is within Windows. Always restart (i.e. reboot) your computer after uninstalling a large application or program.

❖ **Voice over IP (VoIP)**

The practice of using an Internet connection to pass voice data using IP instead of using the standard public switched telephone network. This allows a remote worker to function as if he or

she were directly connected to a PBX even while at home or in a remote office. As well, it skips standard long distance charges, as the only connection is through an ISP. VoIP is being used more and more to keep corporate telephone costs down, as you can simply run two network cables to a desk instead of separate network and data cables. VoIP runs right over your standard network infrastructure, but it also demands a very well-configured network to run smoothly.

- **Webmaster**

A blanket term that refers to the person responsible for running a website. It is associated more with server administration and HTML coding than other Web development, such as CGI scripting.

- **Website**

This term describes a particular company's, user's, or organization's Web pages served up by a webserver. It may be split across multiple servers or URLs, but it is one group of HTML pages with a particular association. For example, the Geek.com website spans multiple servers and includes the domain names ChipGeek, PDAGeek, and Geek.com.

- **General Packet Radio Service (GPRS)**

A mobile phone 2.5G technology that allows short bursts of data between a phone and a network such as the Internet. The short transmissions allow for less costly connectivity than a constant connection, but work best with specialized services designed for such types of communication.

- **Online**

A computer is working online when it is connected via a phone line to another computer or computers on the Internet and is performing any operation such as browsing, downloading or transferring of files or data, etc.

- **Online Banking**

This has become increasingly popular and is yet another way to conduct your account with your bank by using your computer, but Be Warned: Always look for the Secure Site Symbol (a gold padlock) which is usually displayed at the bottom of your screen along with a text message that tells you that you're in a secure zone - before entering your credit card details!

- **Option**

This is one of the few Internet terms that's similar to it's dictionary description i.e. your right to choose. Every time you open up a menu you choose an option which activates or completes a task. Sometimes you'll see a keyboard shortcut indicated alongside the text for that option. Other times you'll see a toolbar icon displayed. This is the computer's way of informing you that if you click on the appropriate toolbar icon the same option will be selected.

- **Object-oriented Programming (OOP)**

This term refers to programming languages that allow you to work with objects. These objects can contain not only data type and data structure information, but also information about how the object can be used by procedures.

❖ **OLE (Object Linking and Embedding)**

A standard for sharing data between applications. It has been around since Windows 3.1 and continues to improve. For example, if you cut a picture out of Paint and paste it into a word processor document, you are using OLE to properly put the data into your document. Of course, if it doesn't work quite right you can blame OLE, or the program's use of it. OLE allows objects to be linked to and embedded in other documents. Linking creates a link to the actual object; embedding puts a copy of the object into the document. You can usually access the program an object was created with in order to edit the linked or embedded object just by clicking on the object. This is much more advanced than just taking a screenshot of the data you want and pasting it into another program as a graphic that has no relation to the original data.

❖ **OLTP (OnLine Transaction Processing)**

A group of programs that allow real-time inputting, recording, and retrieval of data to/from a networked system. The speed of recording the data is critical, as is the reaction time of the system, so that people entering data are not bogged down.

❖ **OSI Model (Open Systems Interconnection Model)**

A way of representing the complexities of computer networking in a 7-layer model, ranging from the physical hardware of networking all the way up to how application programs talk to the network. The 7 layers are: physical, data link, network, transport, session, presentation, and application. The 7-layer OSI model can be used to help diagnose network problems. It is also used as a measurement of how well people know their networking. If you're looking for a job in networking, you should familiarize yourself with the OSI model.

❖ **Output**

Anything that comes out of a computer or system, either electronically or physically, as in, "Watch out for that pile of OUTPUT."

❖ **Pascal**

A programming language that was designed to teach computer science students the concepts of programming. It's almost like C for dummies. In fact, the two languages are very similar, but the stoic brackets of C are changed to happy and fluffy "Begin" and "End" statements. Also, many of the more complex functions of C are missing, but Pascal is still a fully functional language. Borland used Pascal as the basis for its Delphi programming language.

❖ **Peripheral**

Any device that is not part of the motherboard, aside from memory and the CPU. For example, video cards, sound cards, modems, and hard drives are peripherals. When speaking about the exterior of a PC peripheral refers to anything that can be connected to the exterior of a PC, like an LCD monitor or FireWire hard drive.

❖ **Personal Digital Assistant (PDA)**

APDAs are basically beefed-up organizers or toned-down laptops. They have small LCD screens, some form of pen or keyboard input, and software for organization, OCR, and contact management. Some have even more features, such as Web browsing or Internet e-mail. Most PDAs are designed to fit in one hand while you use the other hand for input.

- **Plug-ins**

These are small software programs that extend the capabilities of your Web browser by adding such things as audio, video or animation powers which can turn your computer into a TV or allow you to listen to live radio programs. If you click on a Web page and a banner automatically pops up saying something like " This page works better with Flash" (or Quick-Time Player etc) and it's followed by an invitation to download the relevant plug-in, you can decide if it would be an advantage to do so. Most of them are FREE! If you're not interested, simply close the message box. Plug-ins such as Macromedia Shockwave and RealAudio Player are just two examples that will be required if you want quality audio or video playback on your PC.

- **Pentium II**

This is an improved version of Intel's Pentium Processor which has now been superseded by both the Pentium III and IV. It is likely that it will be bettered many times over the coming years. Computer hardware designers are constantly striving to invent ever faster processors so that their companies can stay ahead of the competition in this cut business.

- **Presentation Program**

Microsoft PowerPoint is the most commonly used Presentation Program at the moment and comes as part of the Microsoft Office suite of applications. It allows you to display slides, show a report, demonstrate a product or service by using visual aids, and generally support a speech or presentation to an assembled audience that may not necessarily be in the same building as the equipment being used.

- **Rambus DRAM (Direct Rambus DRAM)**

One possible future successor to SDRAM, and competitor to DDR SDRAM. DRDRAM (also now simply called RDRAM) was originally developed by Rambus, Inc. This RAM architecture is capable of speeds starting at 266, 356, and 400MHz, and transfers data on the up and down ticks of a clock cycle, where standard SDRAM tops out at about 200MHz. You may see RDRAM referred to at 532, 712, and 800MHz, but that is actually 266*2, 356*2, and 400MHz*2, where the *2 denotes transfer on the up and down tick of the clock. Intel picked RDRAM to be the future of RAM for PCs, but RDRAM ran into many stumbling blocks due to the complexity of manufacturing it. RDRAM was used in the popular Nintendo 64 game machine for its high-speed capabilities and low pin count, necessary for the intense 3D rendering being done on a low complexity system. RDRAM started out using a 16- bit bus, while SDRAM and DDR have 64-bit buses. Thus, 100MHz SDRAM has half the throughput of RDRAM at 400MHz (100*64 = 6,400Mbits/second; 400*2*16 = 12,800Mbits/second). However, newer RDRAM uses a 32-bit bus and is clocked up to 533MHz*2. Still, it never achieved dominance in the PC market.

❖ **Supercomputer**

A computer that is able to operate at a speed that places it at or near the top speed of currently produced computers. Most supercomputers cost millions of dollars, and the traditional model of using one large computer with proprietary hardware is being challenged by using a cluster of cheaper computers with more standard hardware.

❖ **Thin Client**

A thin client is similar to a dumb terminal in that it gets all of its information from the network. Some thin clients have their own memory, but lack a hard drive. They're basically stripped down computers that are supposed to lower the total cost of owning a computer. These computers are generally used in business or commercial applications.

❖ **HAL (Hardware Abstraction Layer)**

The portion of the operating system that lets programs deal with hardware directly. This allows programs needing more speed from the computer to bypass the standard OS calls to hardware. So, instead of calling the Windows' API to draw a dialog box, you would go through the pain of actually drawing it out line by line--that's much faster, but much harder to write. This is of particular importance when coding games.

❖ **L3 Cache (Level 3 Cache)**

This type of cache is becoming more prevalent as microprocessor manufacturers ship more processors with L1 and L2 cache built into the processor. L3 cache is then the extra cache that sits on the motherboard between the processor and main memory, since the processor already contains L1 and L2 cache. Some processors are starting to ship with L3 cache built-in as well to speed up memory operations further. In those cases the L3 cache often sits on a separate area of the die, not built directly into the chip core.

❖ **MAN (Metropolitan Area Network)**

A collection of several LANs that are linked by faster backbone connections. A MAN is smaller than a WAN, and an example is a college campus with several LANs linked on the same network.

❖ **Microcomputer**

The older term for a common home computer, or single processor computer. The next step up is a workstation.

❖ **Microprocessor (MPU)**

This term is commonly used to describe the CPU. More specifically, it refers to the part of the CPU that actually does the work, since many CPUs now contain L1 and L2 caches on-chip.

❖ **Millions of Instructions Per Second (MIPS)**

A rating of how quickly a processor can process instructions. A processor rated at 100 MIPS can process 100,000,000 instructions per second. Although it seems straightforward enough, for an

accurate comparison you must define which instruction(s) you are going to run, and a rating such as this could lead processor makers to build chips that have high MIPS ratings but aren't useful in the real world.

- **Operating System (OS)**

The program that allows you to access the basic functions of your computer. It is the minimum software required to run a program. Thus, you must have an operating system installed and then the programs run on top of it--unless you have a program with an entire operating system embedded into it, which is possible.

- **QA (Quality Assurance)**

The practice of checking hardware, software, or systems for defects, identifying such defects, and then checking to make sure that such defects are corrected when future revisions of software or hardware are ready for testing. QA workers typically work closely with the people who develop hardware and software, and often program exhaustive scripts to automate checking and identify problems.

- **Queue**

A data construct that is first-in, first-out (FIFO). Think of a check-out line at a supermarket, or any type of line formed by people in society. Queues are used throughout the architecture of computers and are necessary in programming languages to accomplish certain tasks. See also stack.

- **W3C (World Wide Web Consortium)**

An industry group created to design and promote standards to increase the functionality of the Web. The W3C was initially established in collaboration with CERN, the creator of the World Wide Web.

- **WAP (Wireless Application Protocol)**

A proposed standard that allows for transfer of data securely between wireless devices, such as PDAs, cellphones, pagers, or other combinations of those devices. WAP supports many different wireless networks.

- **Web Developer**

Grown up Webmasters are Web Developers. Generally, if you are a Web Developer you have a range of Web skills, from managing a webserver to coding HTML, CGI scripts, and even creating spiffy graphics as needed.

- **Webserver**

A machine, or software on a machine, that replies to data requests from a browser using the WWW protocol called HTTP, allowing people to access HTML files.

- **Windows**

A rectangular shaped area on the screen of any GUI that contains a program. It also refers to a family of operating systems developed by Microsoft. The family includes Windows 3.x, 9x, Me, NT, 2000, XP, and 2003.

- **Windows 2000**

At first this operating system was called Windows NT 5, until Microsoft renamed it to Windows 2000. Windows 2000 was mainly a 32-bit operating system using the NT code base, but 64-bit versions also came out for Intel's Itanium processors. Windows 2000 ads new functionality into Windows NT, such as support for USB and other new devices, built-in DirectX 7.0, and many other features.

- **Windows 3.x**

The first accepted commercial version of Windows. There was a version 2.x that was available at one point, but Windows didn't achieve success until 3.x arrived, putting a Windows GUI directly on top of DOS 5.

- **Windows 95**

A 16- and 32-bit operating system that builds on DOS and Windows 3.x to support 32-bit operations, additional hardware, and a more usable interface. This was a true breakthrough product for Microsoft, allowing game developers to run games inside of the Windows OS instead of separately in DOS. It was designed for the consumer market.

- **Windows 98**

This operating system was designed by Microsoft and adds additional hardware support and integrated browsing onto the Windows 95 platform. This OS is still partially 16-bit, a full 15 years after Intel released its first 32-bit processor, the 386.

- **Windows Me (Windows Millennium Edition)**

This Microsoft Windows operating system added many UI enhancements and clutter that actually made it less stable and more hated than Windows 95 and 98. No improvements to the core were made for stability or to enhance the combination 16- and 32-bit mess under the hood. This was Microsoft's last version of Windows containing pieces of 16-bit DOS guts, and good riddance.

- **Windows NT (Windows New Technology)**

A full 32-bit operating system developed by Microsoft to be a very stable operating system to be used on servers and business machines. It was developed from the ground up to be fully 32-bit without much worry about DOS compatibility. The core of Windows NT code has been updated for Windows 2000 and XP.

- **Windows XP**

The friendly-faced, updated version of Windows 2000, with an almost cartoonish interface that will surely be looked back upon with a wince. XP started to take real "advantage" of the Internet by including numerous hooks and links to Microsoft's website to improve various functionality,

and that is part of what made this operating system so controversial. Also, at long last, it moved the Windows NT code-base onto consumer machines, allowing home users to get the benefits of stability that Windows Me didn't offer. Many hold-outs prefer to stick with Windows 2000.

- **Wizard**

An "enhancement" to programs that makes them easier to operate by guiding you through a process, step by step. Often wizards are scoffed at by experienced users, who prefer to do things the hard way.

- **WML (Wireless Markup Language)**

Part of the Wireless Application Protocol (WAP) , allowing text portions of Web content to be separated from graphical content for display on wireless devices.

- **Word**

A group of bits of data regarded as a whole while programming or transferring data. Often a word is 8 bits in length, also referred to as a byte. This is also the name of Microsoft's word processor.

- **Word Size**

The number of bits of data stored in a CPU register. Typically the number is a power of 2, with 8, 16, 32, and 64 being common. You have to deal with word size when doing certain data manipulations while programming.

- **Workstation**

A high-powered computer, one step below a minicomputer and a step above a microcomputer. The term often refers to fairly powerful dual-processor computers used to generate 3D images or manipulate 2D images or sound. Often workstations require very powerful graphics setups.

- **World Wide Web (WWW or Web)**

This is basically a means of communicating text, graphics, and other multimedia objects over the Internet. Web servers on the Internet are set to respond to particular requests sent on TCP/IP port 80 by sending HTML documents to the requester. The requester usually uses a browser to receive this data. Think of the Internet as a 100-lane highway, and the Web as one of those lanes. Of course traffic in the Web lane is probably very high compared to traffic in most other lanes.

- **Worm**

A program that spreads or copies itself from one computer to another without user intervention. Compare this to a virus that only spreads between files on a single computer. Viruses are small-time compared to the havoc that worms can create. Nowadays most malicious programs that you hear about are actually worms, even though software combating them still refers to itself as an anti-virus program. WORM

- **WYSIWYG ("What You See Is What You Get")**

This term describes software that allows you to perform edits on a document that show you on-screen exactly what will print out. The Macintosh was one of the first computers to easily support WYSIWYG programs--that's why it was so popular in desktop publishing. Nowadays WYSIWYG programs are handy when editing HTML pages for display on the Web.

- **Web Browser**

This is the name for the program that is used to navigate the Internet and is used to display web pages on your computer. The majority of people in the world use Microsoft's Internet Explorer Browser. The rest use Netscape Navigator.

- **x86**

The 32-bit instruction set used by most PCs. It originated with the Intel spec of the 8088, which then went to the 8086. Since then increments have come by changing the first number to 80286, 80386, "486", and then Pentium and beyond. The reason that the newer chips are actually named is because the numbers can't be copyrighted, and Intel needed to distinguish its chips from competitors' products.

- **XOR**

An operation that can be executed on two or more binary strings. XOR returns true, or "1", if only one of the two strings contains a 1 at a particular bit position, and a false, or "0", if both strings contain 0s or 1s at both positions. It is similar to the behavior of OR, but is false when both bits are positive (thus "exclusive" one or the other, but not both). For example (0 XOR 0) = 0, (0 XOR 1) = 1, (1 XOR 0) = 1, (1 XOR 1) = 0. Thus: (0011 OR 1001) = 1010.

- **SPARC**

SPARC (Scalable Processor Architecture) is a 32- and 64-bit microprocessor architecture from Sun Microsystems that is based on Reduced Instruction Set Computing (RISC).

For Files and Groups of Maha Malik:

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